# Spectroscopy as a Probe of Spacetime Foam

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Quantum-gravitational spacetime foam varying in space and in time imprints, respectively, a transverse phase decoherence and an achromatic, stochastic arrival-time jitter on light from distant sources. ~~Quantum-gravitational spacetime foam imprints an achromatic, stochastic arrival-time jitter on light from distant sources.~~ We show that the~~this~~ jitter modifies the first-order temporal coherence $g^{(1)}(\tau)$ and hence the observed spectrum, providing a spectroscopic probe of the temporal variation, complementary to existing imaging-channel constraints on the spatial variation. In the dynamical-foam limit, where the foam decorrelates within the light's coherence time, the spectrum splits into a coherent line core suppressed by $e^{-\sigma_\phi^2}$, with $\sigma_\phi^2$ the variance of the foam-induced phase jitter, and an incoherent pedestal, yielding a spectral shape-distortion bound on the foam exponent~~Ng–Perlman foam exponent~~ $\alpha$. Applied to the MeV emission lines of GRB 221009A we obtain $\alpha \gtrsim 0.63$, a weak, lever-arm-dominated floor. For the broadband CMB the foam induces an $e^{-\sigma_\phi^2(\nu)}$ deficit ($\sigma_\phi^2 \propto \nu^2$); a fit to the COBE/FIRAS monopole spectrum reaches $\alpha \gtrsim 0.52$ (95% CL)~~ under the physical Planck-scale pedestal kernel~~, and next-generation spectrometers reach only $\alpha \gtrsim 0.54$–$0.55$ (forecast), limited by the foam–temperature degeneracy. The pedestal-independent line-width and CMB bounds are floors below the holographic value $\alpha = 2/3$, consistent with and excluding no part of the imaging window reported for GRB 221009A; only the conditional TeV survival bound ($\alpha \gtrsim 0.74$, for near-Planck-refresh foam, $\tau_{\rm foam} \lesssim 4 \times 10^{-28}$ s) would exclude it, and set against that window it instead bounds $\tau_{\rm foam}$ from below.~~The pedestal-independent line-width and CMB bounds do not reach the holographic value $\alpha = 2/3$; only the conditional TeV survival bound ($\alpha \gtrsim 0.74$, for near-Planck-refresh foam, $\tau_{\rm foam} \lesssim 4 \times 10^{-28}$ s) would exclude it.~~ All bounds assume foam that decorrelates within the light's coherence time; they vanish in the quasi-static or long-temporal-correlation limits.

## I. INTRODUCTION

A long-standing expectation from the union of quantum mechanics and general relativity is that the smooth pseudo-Riemannian manifold of classical spacetime breaks down at the Planck length $\ell_{\rm P} = \sqrt{\hbar G/c^3} \simeq 1.616 \times 10^{-35}$ m. In Wheeler's original "spacetime foam" picture [1], quantum fluctuations of the metric on this scale render the geometry stochastic; similar conclusions are reached in the Euclidean path-integral treatment [2] and in modern holographic and entropic-gravity arguments from very different starting points. While a complete theory of quantum gravity remains elusive, the question of whether residual imprints of Planck-scale geometric fluctuations can accumulate over cosmological baselines and become observable is one of the few empirical handles on physics in the Planck regime, and it has motivated a substantial body of phenomenological work over the past two decades [3, 4].

A widely studied parametrization due to Ng and collaborators [4–6], the $\alpha$-foam model, encodes the foam dynamics in a power-law path-length variance

$$\sigma_\ell^2(D) = A_\alpha \, D^{2(1-\alpha)} \, \ell_{\rm P}^{2\alpha}, \tag{1}$$

where $D$ is the line-of-sight comoving distance to the source — the appropriate measure rather than the luminosity distance, since the phase accumulation is a property of spacetime itself rather than the photon path [4] — $\alpha$ parametrizes the foam model, and $A_\alpha$ is a dimensionless coefficient of order unity. Ng and Perlman give the foam

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fluctuation equivalently in time-interval form, $\delta t \gtrsim t^{1-\alpha}\, t_{\rm P}^{\alpha}$ with Planck time $t_{\rm P} = \ell_{\rm P}/c$ [4]; since the observable in this work is temporal, that is the more directly relevant statement, recovered explicitly in Eq. (3) below. A photon traversing distance $D$ passes through approximately $N \sim D/\ell_{\rm P}$ independent Planck-scale spacetime cells, each imparting a tiny random displacement to the photon path; the exponent $\alpha$ controls how the per-cell displacements accumulate: for random-walk foam ($\alpha = 1/2$), contributions add in quadrature ($\sigma_\ell \propto \sqrt{D\ell_{\rm P}}$); holographic foam ($\alpha = 2/3$) grows more slowly, motivated by the heuristic that the information in a spatial region scales with its bounding area rather than its volume — an analogy to the holographic entropy bound; and the weak-foam limit ($\alpha = 1$) yields a distance-independent $\sigma_\ell \propto \ell_{\rm P}$, corresponding to perfectly correlated cell contributions.

The primary observational probe of stochastic foam to date has been the imaging channel. Light propagating through spacetime foam accumulates a phase dispersion $\Delta\phi \simeq 2\pi D^{1-\alpha}\ell_{\rm P}^{\alpha}/\lambda$~~$\Delta\phi \simeq 2\pi\ell^{1-\alpha}\ell_{\rm P}^{\alpha}/\lambda$~~ across the entrance aperture, degrading PSF sharpness and erasing interference fringes once $\Delta\phi \gtrsim 2\pi$ [7]. For PKS 1413+135 ($D = 1.216$ Gpc) observed with HST at $\lambda = 1.6\,\mu$m, the random-walk model ($\alpha = 1/2$) predicts $\Delta\phi \sim 10 \times 2\pi$, so the detection of Airy rings in this image marginally rules out that model; the holographic model ($\alpha = 2/3$) predicts $\Delta\phi \sim 10^{-9} \times 2\pi$ and is obviously not ruled out [4, 8, 9]. Models with long-range transverse correlations evade even the random-walk bound [10].

The most luminous GRB ever recorded, GRB 221009A at $z = 0.151$ [11], has been examined in the imaging channel by Steinbring [12] and Steinbring & Ng [13], both building on that author's earlier treatment of high-energy instrumental response, in which telescopes far from their diffraction limits retain only photons within the instrumental acceptance [14, 15]. Using a multiwavelength average of foam-induced angular broadening — analogous to a seeing disk from atmospheric turbulence — Steinbring [12] showed that the holographic ($\alpha = 2/3$) model predicts an angular widening below $\sim 1^\circ$, consistent with the Fermi and Swift localization data for GRBs at $z \leq 5$ up to peak energies of $\sim 10$ MeV. Steinbring & Ng [13] extended this to the full multiwavelength follow-up of GRB 221009A — from optical and near-infrared through X-rays and the highest-energy gamma-rays ever attributed to a GRB — computing the holographic-foam-widened PSF for real telescopes and finding all data consistent with $\alpha = 2/3$. These imaging-channel analyses show that the holographic model is not excluded by the spatial localization of GRB 221009A. Reference [13], developing the GRB 221009A imaging analysis of Ref. [12], advances a stronger reading of the same data. With $\alpha \leq 0.65$ already excluded by images of distant galaxies and AGNs, those authors bracket the foam-widened PSF between two instrumental scalings: $\alpha = 0.650$, where the blurring is scaled to the instrument's field of view, and $\alpha = 0.735$, the weakest-blurring case, scaled to its nominal resolution and still consistent with the localization of GRB 221009A. The holographic $\alpha = 2/3$ falls at the midpoint of that bracket. Across it they find the model consistent with every band observed for this burst and with archival *Fermi* GRB and AGN samples, reporting that LAT images are not diffraction-limited but scale as foam-induced blurring predicts, and reading the resulting halo as attributable to foam. The present work approaches the same exponent through a different channel; Section V A 2 sets its bounds against that bracket.

We develop a framework based on the first-order temporal coherence function $g^{(1)}(\tau)$ of light from astrophysical sources. This function is complementary to the first-order spatial coherence function $g^{(1)}(B)$, which forms the basis of imaging-channel probes. The first-order coherence — the field–field autocorrelation — inherits a foam suppression factor that decreases continuously from unity toward complete fringe erasure as the accumulated phase variance $\sigma_\phi^2$ grows. In our framework the phase variance arises from temporal fluctuations of foam, whereas in the imaging channel it arises from spatial fluctuations; correspondingly, foam imprints itself on the spectrum rather than on the spatial brightness profile.

In our analysis, the foam acts on the electric field: foam modulates the first-order temporal coherence $g^{(1)}(\tau)$ along a fixed line of sight, and the observable effect is a spectral shape distortion. Foam varying in both space and time will scatter light in angle as well as distort it in frequency; the angular channel is the imaging literature reviewed above. This article focuses on spectral distortion and does not account for any accompanying deflection: the GRB bound uses the *fractional* line width, a ratio in which any flux lost from the beam cancels; the CMB bound uses the all-sky monopole, which angular scattering redistributes among sky directions but cannot deplete. The exception is the TeV survival probe, which turns on whether broadband flux survives at all and so cannot distinguish flux lost to deflection from flux removed spectrally. Source extension is likewise immaterial, an incoherent source being the intensity-weighted mean of single-sightline results. Appendix A establishes both the source-extension result and the insensitivity to deflection.

The spectroscopic channel has been considered previously. Cooke *et al.* [16] postulate a convolution kernel, a heuristic that the authors note has not been demonstrated by a model of spacetime quantization. Their bound comes from a narrow Fe II absorption line from a quiescent cloud at $z \simeq 2.34$ with VLT/ESPRESSO which, adopting the same exponent through $\sigma_E/E = (E/E_{\rm P})^\alpha$, ruled out $\alpha \leq 0.634$. What distinguishes our work is the physical assumptions and the details of the predictions. We derive the distortion from the first-order coherence of the electric field propagating through a fluctuating metric — the light-cone fluctuations of Refs. [17, 18]. The two models differ in their predicted effect on spectra. The Cooke *et al.* kernel is a Gaussian of width comparable to the line, so the

displaced flux stays in the line. In this article, flux is redistributed into a core $e^{-\sigma_\phi^2} S_{\rm src}(\nu)$ retaining the source profile — Gaussian for the line considered here — at reduced weight, and a pedestal component of width $\sim 1/\tau_{\rm foam}$ carrying the complementary $1 - e^{-\sigma_\phi^2}$. The in-band observable is therefore a flux deficit, $1 - e^{-\sigma_\phi^2(\nu)}$ with $\sigma_\phi^2 \propto \nu^2$. Within the core the absolute width in fact narrows, so broadening survives only in the fractional width and only at second order, $r_{\rm foam} \propto r_{\rm int}^2\, \sigma_\phi$. Our derivation hinges on the delay varying within the light's coherence time; in the quasi-static limit the effect vanishes entirely, a regime on which the model of Ref. [16], having no time dependence, is silent.

Because the foam variance enters the resulting bounds only through the doubly-logarithmic factor $\ln(D_C/\ell_P) \approx 138$–$141$ (GRB to CMB), the bounds are set primarily by source geometry and probe energy; the data enter as a small logarithmic correction (a decade of measurement precision shifts $\alpha$ by $\lesssim 0.02$), so they are robust but cannot be appreciably tightened by better data of the same kind.

The remainder of the paper is organized as follows. Section II introduces the spacetime foam model: the arrival-time jitter parameterization, the temporal autocorrelation $C_\alpha(\tau)$, and the distribution of differential jitter. Section III derives the first-order coherence function $g^{(1)}_{\rm obs}(\tau)$ of the observed field and establishes the Gaussian closed form. Section IV translates these results into spectroscopic observables via the Wiener–Khinchin theorem, establishing the split of the observed spectrum into a coherent line core suppressed by $e^{-\sigma_\phi^2}$ and a broad pedestal, and connects them to the imaging Strehl ratio. Section V applies the spectral shape-distortion bound to two sources: GRB 221009A (Section V A) and the CMB as observed by COBE/FIRAS (Section V B). Section VI summarizes the main results and outlines future observational directions. Table III collects the principal symbols used throughout.

## II. SPACETIME FOAM MODEL

### A. Arrival-Time Jitter

We adopt the Ng–Perlman picture of spacetime foam: stochastic, achromatic path-length fluctuations whose cumulative effect along a line of sight is an arrival-time jitter $\delta t(t)$, which a carrier reads as a phase jitter $\sigma_\phi$. Following the prior foam literature we take a point source on a single line of sight. For an incoherent extended source the observed coherence is the intensity-weighted mean of such points: cross terms between distinct points vanish in $\langle E^* E\rangle$, so no foam correlation between sightlines enters (Appendix A).~~; at the level of first-order coherence an extended incoherent source is the linear superposition of such points. The transverse behavior of the foam across an extended source is beyond our scope.~~

We model the detected field as a stochastically time-warped version of the intrinsic source field:

$$E_{\rm obs}(t) = E_{\rm src}(t - \delta t(t))\,, \tag{2}$$

where $E_{\rm src}(t)$ is a single mode of a stationary, zero-mean complex Gaussian (chaotic) field — appropriate for the thermal and incoherent sources here (GRB high-energy emission, CMB) — and $\delta t(t)$ is the foam jitter, a stationary process independent of $E_{\rm src}$. Equation (2) warps only the time argument of the field on a fixed line of sight, and the ensemble-mean intensity is preserved, $\langle |E_{\rm obs}|^2\rangle = \langle |E_{\rm src}|^2\rangle$. Here $t$ is the observer's time, with the deterministic light-travel delay and cosmological time dilation absorbed into the argument of $E_{\rm src}$, so $\delta t(t)$ is the zero-mean foam fluctuation about the mean arrival time. The jitter arises from cumulative path-length fluctuations along the line of sight of propagation distance $D$:

$$\delta t(t) = \frac{1}{c}\,\Delta\ell_{\rm tot}(t). \tag{3}$$

Equivalently, the rms jitter $\sigma_{\delta t} = \sigma_\ell(D)/c = \sqrt{A_\alpha}\, t^{1-\alpha} t_{\rm P}^{\alpha}$, with light-travel time $t = D/c$ and Planck time $t_{\rm P} = \ell_{\rm P}/c$, is exactly the time-interval fluctuation of Ng and Perlman [4] — the temporal form on which the present analysis rests.

This form is motivated by results on light-cone fluctuations. Ford [17] showed that metric fluctuations from a graviton bath smear the light-cone singularity of the field's two-point functions, the metric-averaged retarded Green's function acquiring Gaussian support both inside and outside the classical light cone — some photons arriving early, others late. Ford and Svaiter [18] carried this to the two-time level, showing that flight times are Gaussian-distributed about the classical value and that photons emitted in rapid succession have correlated flight times, for which they derive a correlation function. Equation (2), with a stationary Gaussian $\delta t(t)$ of correlation $C_\alpha(\tau)$, is the field-level encapsulation of those two results.

Because the path-length fluctuation is geometric—arising from stochastic metric perturbations of the FLRW background rather than a dispersive medium—$\delta t$ is achromatic (frequency-independent), distinguishing foam from plasma

dispersion, interstellar scintillation, and standard spectral distortions [4]. A monochromatic carrier of observed frequency $\omega_0 = 2\pi\nu_0$ reads the same delay as a phase jitter $\delta\phi = \omega_0\,\delta t$, giving a phase variance

$$\sigma_\phi^2 \;=\; \left(\frac{2\pi}{\lambda_{\rm obs}}\right)^2 \sigma_\ell^2(D_C). \tag{4}$$

This is the phase accumulation along a single ray, or in a transverse patch in which the foam is coherent. ~~on which all spectroscopic bounds below rest.~~

Chen *et al.* [19] point out that the phase dispersion is damped relative to the path-length dispersion for light propagating through a volume of incoherent foam. The field samples a three-dimensional region rather than following a ray, and transverse structure finer than $\lambda$ scatters only into evanescent modes, so that for cells uncorrelated beyond $\ell_{\rm P}$ the accumulated phase amplitude carries an extra factor $\ell_{\rm P}/\lambda$ — some $5\times10^{-22}$ at 37 MeV, and its square in the variance. The effect thus depends on the transverse correlation length of the foam, which Eq. (1) does not fix and which we do not model here. Lee *et al.* [10] work through a specific case, the pixellon realization of the Verlinde–Zurek effect, in which holographic degrees of freedom on the boundary of a causal diamond drive fluctuations in its size; the accumulated variance matches the random walk, $\langle\Delta L^2\rangle \sim \ell_{\rm P} D$, but the power sits in the low angular harmonics rather than at $1/\ell_{\rm P}$. They find the suppression absent for such fluctuations, whose power lies at wavenumbers the field can carry. We adopt Eq. (4) as a premise, shared with the imaging constraints and with the $\alpha$-foam literature generally. It is worth noting that the transverse correlation structure bears on a purely temporal observable: the filtering acts on $\sigma_\phi^2$ itself rather than on its transverse gradient, and so reaches the spectrum as directly as it reaches the image.

### B. Temporal Decorrelation of Spacetime Foam

The Ng–Perlman variance $\sigma_\ell^2(D)$ (Eq. (1)) fixes the amplitude of the path-length fluctuations along a single ray but not their temporal correlation. Quantum-foam models posit that each Planck-scale cell fluctuates on a characteristic timescale $\tau_{\rm foam}$, so the jitter at times $t$ and $t+\tau$ is correlated for $\tau \ll \tau_{\rm foam}$ and essentially independent for $\tau \gg \tau_{\rm foam}$. We encode this in the normalized temporal autocorrelation $C_\alpha(\tau; D)$ of the integrated path-length fluctuation $\Delta\ell_{\rm tot}(t)$,

$$\big\langle \Delta\ell_{\rm tot}(t)\,\Delta\ell_{\rm tot}(t+\tau)\big\rangle = \sigma_\ell^2(D)\,C_\alpha(\tau; D), \tag{5}$$

normalized so that $C_\alpha(0; D) = 1$. Because $C_\alpha$ is defined on the integrated path, its temporal shape inherits a residual $D$-dependence from the way the per-cell foam dynamics over the $\sim D/\ell_{\rm P}$ cells crossed by the line of sight combine; the explicit $D$-argument records this dependence.

The underlying foam correlation is spatio-temporal, $C(\mathbf{r}, \tau)$, and $C_\alpha(\tau; D)$ is its effective temporal part along the ray. The angular extent of the sources does not enter, since for an incoherent source the sightlines superpose in intensity and the foam is never averaged across them (Appendix A). ~~Since no physical source is pointlike, the underlying foam correlation is spatio-temporal, $C(\mathbf{r},\tau)$, and we read $C_\alpha(\tau; D)$ as the effective temporal correlation left after coarse-graining over the source's emitting four-volume and propagation path.~~ The reduction to a purely temporal $C_\alpha$ assumes the spatio-temporal correlation factorizes, $C(\mathbf{r},\tau) \approx C_{\rm sp}(\mathbf{r})\,C_\alpha(\tau)$, with the spatial factor absorbed into the normalization; we flag this as an assumption, since the spatial correlation structure over cosmological path lengths is not constrained here. Non-factorization would modify the combination along the path and hence the pedestal-kernel shape; the core suppression $e^{-\sigma_\phi^2}$ involves $C_\alpha$ only through its long-lag limit $C_\alpha \to 0$ and is untouched. ~~Non-factorization would modify the spatial averaging and hence the pedestal-kernel shape; the in-band observables (the core suppression $e^{-\sigma_\phi^2}$) depend on the temporal $C_\alpha$ along a fixed line of sight and are less sensitive to it.~~

The timescale $\tau_{\rm foam}$ is an additional free parameter of the foam model, not fixed by the $\alpha$-model variance relation alone. In the standard phenomenological framework [6, 20], the foam refreshes on the Planck time $\tau_P \equiv \ell_P/c \approx 5\times10^{-44}$ s, so the integrated path-length jitter $\Delta\ell_{\rm tot}(t)$ — a sum of $\sim D/\ell_P$ independent contributions — likewise decorrelates on timescale $\tau_{\rm foam} \sim \tau_P$. This is the dynamical-foam regime adopted throughout: $\tau_{\rm foam}$ is far shorter than any timescale on which any observable coherence has support, so $C_\alpha(\tau) \to 0$ there. By contrast, models with long-range temporal correlations — the temporal analogue of the long-range transverse spatial correlations of Ref. [10] that evade interferometric imaging bounds — would give $C_\alpha(\tau) \approx 1$ over macroscopic lags and drive $\sigma_\Delta^2(\tau) \to 0$, suppressing the foam signal; the bounds derived below therefore rest on the dynamical-foam assumption.

### C. Differential Jitter and Its Distribution

We now define the differential jitter, which enters the coherence integrals, between photons arriving at times $t$ and $t+\tau$:

$$\Delta(\tau) \equiv \delta t(t+\tau) - \delta t(t). \tag{6}$$

Expanding the square and applying Eq. (5):

$$\begin{aligned}\sigma_\Delta^2(\tau) &= \frac{1}{c^2}\left\langle [\Delta\ell_{\rm tot}(t+\tau) - \Delta\ell_{\rm tot}(t)]^2 \right\rangle \\ &= \frac{1}{c^2}\left[\sigma_\ell^2 + \sigma_\ell^2 - 2\sigma_\ell^2\, C_\alpha(\tau)\right] \\ &= \frac{2\sigma_\ell^2(D)}{c^2}\left[1 - C_\alpha(\tau; D)\right]. \end{aligned} \tag{7}$$

By the central limit theorem (CLT), $\Delta(\tau)$ is a sum of contributions from many foam cells and is therefore Gaussian with zero mean:

$$\Delta(\tau) \sim \mathcal{N}\big(0,\, \sigma_\Delta^2(\tau)\big)\,, \tag{8}$$

where $\sigma_\Delta^2$ is taken from Eq. (7); for any source at $D \gtrsim 1$ kpc, the cell count $N_{\rm cells} \sim D/\ell_{\rm P} \gtrsim 10^{54}$ amply satisfies the large-$N$ requirement and higher-cumulant corrections are negligible. For $\alpha = 1/2$ (random walk), per-cell path-length variances are equal ($\ell_{\rm P}^2$ each) and the CLT applies in its standard form. For $\alpha \neq 1/2$, the per-cell contributions scale differently with $D$ and the equality of per-cell variances is a model assumption rather than a direct CLT consequence; and at the correlated ($\alpha \to 1$) end the per-cell increments are not independent, so the large-$N$ argument does not apply. Gaussianity is then adopted as a modeling choice, minimal and consistent with the $\alpha$-model.

## III. FIRST-ORDER COHERENCE

### A. Jitter Convolution Formula for $g^{(1)}$

The observed first-order coherence is

$$g_{\rm obs}^{(1)}(\tau) = \frac{\langle E_{\rm obs}^*(t)\, E_{\rm obs}(t+\tau)\rangle}{\langle |E_{\rm obs}(t)|^2\rangle}. \tag{9}$$

Substituting Eq. (2) and letting $u = t - \delta t(t)$, $v = t + \tau - \delta t(t+\tau)$, so that $v - u = \tau - \Delta(\tau)$, stationarity of $E_{\rm src}$ gives

$$\langle E_{\rm src}^*(u)\, E_{\rm src}(v)\rangle = \langle |E_{\rm src}|^2\rangle\, g_{\rm src}^{(1)}(\tau - \Delta(\tau)). \tag{10}$$

Since $\delta t$ is independent of $E_{\rm src}$ and $\langle |E_{\rm obs}|^2\rangle = \langle |E_{\rm src}|^2\rangle$ by stationarity (so the Eq. (9) normalization cancels), averaging over the jitter yields

$$g_{\rm obs}^{(1)}(\tau) = \big\langle g_{\rm src}^{(1)}(\tau - \Delta(\tau))\big\rangle_{\delta t} = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma_\Delta^2(\tau))\, g_{\rm src}^{(1)}(\tau - \Delta)\, d\Delta. \tag{11}$$

For a quasi-monochromatic source $g_{\rm src}^{(1)}(\tau) = e^{-i\omega_0\tau}\,\gamma(\tau)$ with $\gamma$ the slowly varying envelope coherence, Eq. (11) reads

$$g_{\rm obs}^{(1)}(\tau) = e^{-i\omega_0\tau}\left\langle e^{\,i\omega_0\Delta(\tau)}\,\gamma(\tau - \Delta(\tau))\right\rangle_{\delta t}, \tag{12}$$

the envelope responding to the achromatic differential delay $\Delta(\tau)$ (variance $\sigma_\Delta^2$) and the carrier acquiring the phase $\omega_0\Delta(\tau)$. The carrier suppression is the characteristic function of that phase,

$$\mathcal{F}_{\rm foam}(\tau) \equiv \left\langle e^{\,i\omega_0\Delta(\tau)}\right\rangle_{\delta t} = e^{-\omega_0^2\sigma_\Delta^2(\tau)/2} = e^{-\sigma_\phi^2\,[1 - C_\alpha(\tau; D)]}, \tag{13}$$

with $\mathcal{F}_{\rm foam}(0) = 1$, the differential phase variance $D_\phi(\tau) \equiv \omega_0^2\sigma_\Delta^2(\tau) = 2\sigma_\phi^2[1 - C_\alpha(\tau; D)]$, and $\sigma_\phi^2$ from Eq. (4); at $\tau \to \infty$ ($C_\alpha \to 0$), $\mathcal{F}_{\rm foam} \to e^{-\sigma_\phi^2}$.

### B. Gaussian Source Profile

The GRB 221009A emission line is well described by a Gaussian profile in each epoch [21], and it serves as the representative case admitting the simplest closed form. A Gaussian line of $1\sigma$ width $\sigma_\nu$ centered on $\nu_0$ corresponds via the Wiener–Khinchin theorem to the coherence function

$$g^{(1)}_{\rm src}(\tau) \;=\; e^{-i\omega_0\tau}\, e^{-\tau^2/2\tau_G^2}, \tag{14}$$

where $\omega_0 \equiv 2\pi\nu_0$ and $\tau_G \equiv 1/(2\pi\sigma_\nu)$ is the Gaussian coherence decay time. Substituting the Gaussian profile (14) into the jitter-convolution integral (12) and evaluating the resulting Gaussian integral over $\Delta$ (with $\sigma^2_\Delta \equiv \sigma^2_\Delta(\tau)$) gives the exact closed form

$$\begin{aligned} g^{(1)}_{\rm obs}(\tau) = \frac{\tau_G}{\sqrt{\tau_G^2+\sigma^2_\Delta(\tau)}}\, \exp\!\left[-i\omega_0\tau + \frac{i\,\omega_0\,\sigma^2_\Delta(\tau)\,\tau}{\tau_G^2+\sigma^2_\Delta(\tau)}\right] \\ \times \exp\!\left[-\frac{\tau^2}{2\big[\tau_G^2+\sigma^2_\Delta(\tau)\big]}\right] \exp\!\left[-\frac{D_\phi(\tau)}{2} + \frac{\omega_0^2\sigma^4_\Delta(\tau)}{2\big[\tau_G^2+\sigma^2_\Delta(\tau)\big]}\right]. \end{aligned} \tag{15}$$

The observed coherence is again a Gaussian-modulated oscillation, but with three foam-induced modifications: (i) a slight shift of the carrier, $\nu_0^{\rm eff} = \nu_0\big[1-\sigma^2_\Delta(\tau)/(\tau_G^2+\sigma^2_\Delta(\tau))\big]$; (ii) a wider Gaussian decay of $g^{(1)}_{\rm obs}(\tau)$ in lag (decay scale $\sqrt{\tau_G^2+\sigma^2_\Delta(\tau)}$ vs. $\tau_G$); and (iii) a global amplitude suppression set by the differential phase variance $D_\phi(\tau) = \omega_0^2\sigma^2_\Delta(\tau)$ (Eq. (13)), at the carrier $\omega_0$. The marginal phase variance $D_\phi = \omega_0^2\sigma^2_\Delta$ is exact, while the carrier shift (i) and the $\omega_0^2\sigma^4_\Delta$ amplitude term are carrier–envelope coupling of order $\sigma^2_\Delta/\tau_G^2$. In the slowly-varying-envelope limit $\sigma_\Delta \ll \tau_G$ these vanish and $\mathcal{F}_{\rm foam} = e^{-D_\phi/2}$ factors out cleanly, $g^{(1)}_{\rm obs} \to \mathcal{F}_{\rm foam}(\tau)\, g^{(1)}_{\rm src}(\tau)$ — the pure multiplicative suppression on which the spectroscopic bounds rest. Normalization $g^{(1)}_{\rm obs}(0) = 1$ holds because $\sigma_\Delta(0) = 0$ (hence $D_\phi(0) = 0$). The narrowband limit $\tau_G \to \infty$ of Eq. (15) reduces to a pure phase-modulated oscillation, which is the single-mode building block from which the broadband CMB spectrum of Section V B 2 is constructed by incoherent superposition.

## IV. IMPLICATIONS FOR SPECTROSCOPY

### A. The Wiener–Khinchin Bridge

The foam-modified observed coherence function — given in general by the jitter-convolution integral Eq. (11) with the closed form Eq. (15) — carries direct spectroscopic content: for a stationary field the power spectral density $S(\nu)$ and the field autocorrelation $\langle E^*(t)E(t+\tau)\rangle$ are~~$g^{(1)}(\tau)$ are~~ Fourier-transform pairs, implying

$$S_{\rm obs}(\nu) = \langle |E_{\rm obs}|^2\rangle\, \mathcal{F}\Big\{ g^{(1)}_{\rm obs}(\tau)\Big\}(\nu). \tag{16}$$

Here $\mathcal{F}\{f\}(\nu) \equiv \int_{-\infty}^{\infty} d\tau\, e^{\,i2\pi\nu\tau} f(\tau)$. The observed field is stationary because $E_{\rm src}$ and $\delta t$ are and are independent, so the theorem applies, and the prefactor fixes the normalization: $g^{(1)}(0) = 1$ gives $\int S_{\rm obs}\, d\nu = \langle |E_{\rm obs}|^2\rangle$. ~~The theorem applies because astronomical emission lines have finite linewidth (so the Fourier transform is well defined) and because the foam-induced jitter is a stationary process independent of the source (foam acts on light after emission, while $\delta t(t)$ statistics are time-translation invariant on observational timescales). The observed field is then stationary when the integration time exceeds the foam coherence time $\tau_{\rm foam}$ (the lag scale beyond which $C_\alpha(\tau; D) \to 0$), so that the time-averaged spectrum coincides with its ensemble average: $T_{\rm int} \gg \tau_{\rm foam}$. Because $\tau_{\rm foam} \sim \tau_P \sim 10^{-44}$ s is vastly shorter than the timescale of any astrophysical evolution (e.g. the hours over which the GRB line centroid sweeps 37 $\to$ 6 MeV), the slow kinematic evolution and the fast foam jitter are cleanly separable, and each epoch is stationary on the foam timescale.~~

In the dynamical limit $C_\alpha(\tau; D) \to 0$ for $|\tau| \gtrsim \tau_{\rm foam}$, $\mathcal{F}_{\rm foam}(\tau)$ acquires a long-lag plateau $e^{-\sigma^2_\phi}$ outside a narrow window $|\tau| \lesssim \tau_{\rm foam}$. Fourier-transforming via Eq. (16), the plateau produces a coherent core $e^{-\sigma^2_\phi} S_{\rm src}(\nu)$, while the short-lag window redistributes the complementary fraction $1 - e^{-\sigma^2_\phi}$ of the source flux into a broad pedestal. This core-plus-pedestal split holds for any source, with $\sigma^2_\phi$ evaluated frequency by frequency ($\sigma^2_\phi \to \sigma^2_\phi(\nu)$) for a broadband source; its broadband form for the CMB, including the pedestal shape, is worked out for a specific model of the temporal autocorrelation $C_\alpha$ in Section V B 2. The core-plus-pedestal split is the optical analogue of the Mössbauer recoil-free fraction: Gaussian phase statistics and unit modulus reduce any such modulation to a sharp elastic line of

weight $e^{-\sigma_\phi^2}$, retaining the source profile, plus an inelastic remainder carrying $1-e^{-\sigma_\phi^2}$ spread over the far broader bandwidth set by the perturbation's correlation time.

For the Gaussian source profile of Section III B, Fourier-transforming Eq. (15) with $\sigma_\Delta^2$ held at its asymptotic value $2\sigma_\ell^2/c^2$ gives the explicit Gaussian core

$$S_{\rm obs}^{\rm core}(\nu) \;=\; \frac{\tau_G}{\sqrt{\tau_G^2+\sigma_\Delta^2}}\,\exp\!\left[-\frac{\sigma_\phi^2\,\tau_G^2}{\tau_G^2+\sigma_\Delta^2}\right]\cdot\frac{1}{\sqrt{2\pi}\,\sigma_{\rm obs}}\exp\!\left[-\frac{(\nu-\nu_0^{\rm eff})^2}{2\sigma_{\rm obs}^2}\right], \tag{17}$$

with $\nu_0^{\rm eff}=\nu_0\tau_G^2/(\tau_G^2+\sigma_\Delta^2)$ and $\sigma_{\rm obs}=1/[2\pi\sqrt{\tau_G^2+\sigma_\Delta^2}]<\sigma_\nu$; the result is exact for the single-carrier Gaussian source of Eq. (15). The Gaussian's carrier shift, line narrowing, and peak suppression are the frequency-domain images of items (i)–(iii) of Section III B.

The complementary $1-e^{-\sigma_\phi^2}$ of the line flux reappears as a pedestal: in the regime $\sigma_\Delta \ll \tau_G$, the plateau/residual decomposition of $\mathcal{F}_{\rm foam}$ sends the short-lag residual to the pedestal

$$\begin{aligned} S_{\rm obs}^{\rm ped}(\nu) &= \int_{-\infty}^{\infty} d\tau\, e^{\,i2\pi(\nu-\nu_0)\tau}\, e^{-\tau^2/2\tau_G^2}\left[e^{-\sigma_\phi^2[1-C_\alpha(\tau)]}-e^{-\sigma_\phi^2}\right] \\ &\xrightarrow{\tau_{\rm foam}\ll\tau_G} \hat{R}(\nu-\nu_0)=\tau_{\rm foam}\,\hat{h}\big[(\nu-\nu_0)\,\tau_{\rm foam}\big], \end{aligned} \tag{18}$$

where the limit rescales $\tau=\tau_{\rm foam}x$, $h(x)\equiv e^{-\sigma_\phi^2[1-C_\alpha(\tau_{\rm foam}x)]}-e^{-\sigma_\phi^2}$, and $\hat{h}(y)\equiv\int_{-\infty}^{\infty}dx\,e^{\,i2\pi xy}\,h(x)$ is a fixed $\mathcal{O}(1)$ function of unit-order width; the full observed line is $S_{\rm obs}(\nu)=S_{\rm obs}^{\rm core}(\nu)+S_{\rm obs}^{\rm ped}(\nu)$. The pedestal width $\Delta\nu_{\rm ped}\sim 1/\tau_{\rm foam}$ exceeds the line width $\sigma_\nu=1/(2\pi\tau_G)$ by $\tau_G/\tau_{\rm foam}\gg 1$ (Fourier reciprocity: narrow in lag, broad in frequency), and its total weight $\int S_{\rm obs}^{\rm ped}\,d\nu=1-e^{-\sigma_\phi^2}$ complements the core's $e^{-\sigma_\phi^2}$, conserving the integrated line flux~~conserving line photon number~~. Spread over $\sim 1/\tau_{\rm foam}$ — for Planck-scale foam $\Delta\nu_{\rm ped}\sim 10^{43}$ Hz — the pedestal's in-line contribution is suppressed by $\tau_{\rm foam}/\tau_G\ll 1$, so the spectrum near the line peak is dominated by the coherent core.

The core and pedestal differ in predictive status. The suppression $e^{-\sigma_\phi^2(\nu)}$ is exact for any $\sigma_\phi^2$ — from Gaussian phase statistics and long-lag decorrelation ($C_\alpha\to 0$) — so the linearization $1-e^{-\sigma_\phi^2}\simeq\sigma_\phi^2$ is only a convenience. The observed deficit equals $1-e^{-\sigma_\phi^2(\nu)}$ while the pedestal stays below the core. Being broad ($\sim 1/\tau_{\rm foam}$), the pedestal returns only a fraction $f_{\rm band}\sim\Delta\nu\,\tau_{\rm foam}$ to a band $\Delta\nu$, so the core exceeds this floor for $\sigma_\phi^2\lesssim -\ln f_{\rm band}$ — deep into the optically thick regime, the more so as self-convolution broadens the pedestal. Beyond it the deficit saturates at the floor: optically thick foam remains evident, but its strength requires the pedestal profile, the $C_\alpha$-dependent self-convolution series. Flux conservation~~Photon-number conservation~~ preserves the qualitative signature throughout.

### B. Spectral Analogue of~~Connection to~~ the Imaging Strehl Ratio

The first-order coherence function $g_{\rm obs}^{(1)}(\tau)$ derived above describes temporal coherence along a single line of sight, with the foam suppression governed by the temporal decorrelation $C_\alpha(\tau;D)$ of Eq. (13). The Strehl ratio of an imaging telescope is the spatial counterpart: phase variations across the aperture plane are governed by an aperture-plane coherence function $C_\alpha^{\rm sp}(B;D)$, constructed by direct analogy to the temporal $C_\alpha(\tau;D)$ of Section II B under the same foam-cell statistics.

The structural equivalence is not merely formal: in both cases the foam suppression arises as the characteristic function of a zero-mean Gaussian differential random variable. Temporally, the variable is the differential jitter $\Delta(\tau)=\delta t(t+\tau)-\delta t(t)$ with variance $\sigma_\Delta^2(\tau)=(2\sigma_\ell^2/c^2)[1-C_\alpha(\tau;D)]$ (Eq. (7)). Spatially, the variable is the differential phase $\Delta\phi(B)=\phi(\mathbf{x}+\mathbf{B})-\phi(\mathbf{x})$ between two points on the aperture plane separated by baseline $B$, with variance

$$\sigma_{\Delta\phi}^2(B) \;=\; 2\sigma_\phi^2\big[1-C_\alpha^{\rm sp}(B;D)\big]. \tag{19}$$

The focal-plane field of a telescope is the Fraunhofer Fourier transform of the aperture-plane field, giving the diffraction-limited PSF when the aperture phase is uniform. Foam imprints a Gaussian random phase $\phi(\mathbf{x})$ on the aperture with variance $\sigma_\phi^2$ and two-point correlation $C_\alpha^{\rm sp}(B;D)$, blurring this PSF. In the resolved limit $C_\alpha^{\rm sp}(D_{\rm tel};D)\to 0$ — where aperture phases at distinct points decorrelate — the on-axis peak is attenuated by the Strehl ratio

$$S \;\simeq\; e^{-\sigma_\phi^2} \tag{20}$$

(extended-Maréchal [4]), the surrounding Airy fringes are washed out by the same factor, and in the imaging channel the complementary $1-e^{-\sigma_\phi^2}$ of the source flux is scattered in angle into an incoherent halo — the "halo" of optical

photometry~~the complementary $1 - e^{-\sigma_\phi^2}$ of the source flux scatters into an incoherent halo~~. The pedestal is its spectroscopic counterpart: the same fraction of the flux, displaced in frequency rather than in angle; where the foam is transversely uncorrelated across the aperture, the flux that leaves the image core is the flux that leaves the coherent core (Appendix A).

Imaging and spectroscopic channels each probe $\sigma_\phi^2$ through a Fourier transform — aperture position $\leftrightarrow$ focal-plane angle in imaging (Fraunhofer), lag $\leftrightarrow$ frequency in spectroscopy (Wiener–Khinchin) — with the rapid chromatic tightening $\sigma_\phi \propto 1/\lambda$ exploited in both Ng & Perlman's [4] Table 1 and the analyses of Section V A. The shared $\sigma_\phi^2$ that suppresses the coherent diffraction-limited PSF in imaging also suppresses the coherent line core in spectroscopy, placing the published imaging constraints [4, 7–9] on a common footing with the present framework.

Two caveats temper the duality. A quantitative cross-channel comparison is model-dependent: it requires relating the spatial ($C_\alpha^{\rm sp}$) and temporal ($C_\alpha$) foam correlation functions under a common foam-cell model, so the constraints coincide only to the precision of those shared details. And the channels differ in their reference truth: in imaging the aperture is known a priori, whereas in spectroscopy the intrinsic source spectrum is itself uncertain, making the foam contribution harder to isolate.

## V. SOURCES

Time-varying spacetime foam modifies the source first-order coherence, and hence its spectrum, in a wavelength-dependent way through the scaling $\sigma_\phi \propto 1/\lambda$. Two observable consequences follow. First, spectral features distort per Eq. (17) with a carrier shift, line narrowing, and peak suppression scaling as $\sigma_\phi \propto 1/\lambda$. Although the absolute width narrows (item (ii), Section III B), the center shifts to lower frequency by a larger factor (item (i)), so the fractional width $\sigma/\nu$ broadens — the quantity constrained below. Second, the broadband spectral energy distribution is distorted: the cosmic microwave background, expected to be a near-perfect blackbody, would acquire foam-induced deviations whose chromatic signature is distinct from standard chemical-potential and Compton-$y$ distortions.

### A. GRB 221009A

#### 1. *The source*

GRB 221009A ("the BOAT," Brightest Of All Time) is the most energetic gamma-ray burst in the instrumental record, with isotropic-equivalent energy $E_{\rm iso} \sim 10^{55}\,$erg at redshift $z = 0.151$ [11], a once-in-$\sim 10^4$-year event [22]. Adopting the Planck 2018 flat ΛCDM cosmology used throughout [23] ($\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$, $H_0 = 67.36\,{\rm km\,s^{-1}\,Mpc^{-1}}$), the comoving distance is $D_C(z = 0.151) \approx 0.65\,$Gpc $\approx 2.0 \times 10^{25}\,$m and $\ln(D_C/\ell_P) \approx 138.4$.

Two features of this event make it a unique foam laboratory. First, LHAASO detected more than 5000 gamma rays above 0.5 TeV (WCDA) [24] and 142 events above 3 TeV (KM2A) extending to $\sim$13 TeV [25]; these provide an absolute-flux test of the foam suppression. Second, two independent analyses report a narrow gamma-ray emission feature in the Fermi/GBM data, interpreted as blueshifted electron–positron pair annihilation [11, 21], at observed energies spanning 6–37 MeV. Zhang et al. [21] detect a Gaussian line whose centroid evolves as $E_{\rm line} \propto t^{-1}$ from $\approx 37\,$MeV to $\approx 6\,$MeV, with $t$ measured from the fitted decay onset $t_0 \approx 226\,$s post-trigger, with a fractional width $r_{\rm obs} \equiv \sigma_{\rm obs}/E_{\rm line}$ of weighted mean $0.111 \pm 0.014$, the seven sub-intervals spanning $\approx 0.07$–$0.20$ and consistent with a constant across the 9–37 MeV lever arm ($\chi^2/{\rm dof} = 5.3/6$; Table I); the foam analysis depends only on the Doppler-invariance of the line's fractional width, not on the emission mechanism. The line is resolved. The BGO energy resolution near an MeV is $\approx$10% FWHM, or $\sigma_{\rm LSF}/E \approx 0.04$, well below the measured $r_{\rm obs}$, and the spectral fits of Ref. [21] forward-fold the detector response, so the widths quoted there are already corrected for it. The instrumental line-spread function therefore does not enter the fractional-width bound. Nor does it enter the TeV survival bound, which counts photons rather than measuring a profile.

These two data sets yield two complementary foam probes that draw on the same $\sigma_\phi^2 \propto \nu^2$ physics but differ sharply in strength and in what they assume. The stronger is an absolute bound: the mere survival of TeV photons bounds $\sigma_\phi^2$ directly. The weaker is a relative one: energy-dependent broadening of the MeV line constrains $\alpha$ only in the deep strong-foam regime. Both operate in the dynamical-foam regime $\tau_{\rm foam} \ll \tau_c$ — with $\tau_c \equiv \tau_G = 1/(2\pi\sigma_\nu)$ the coherence time of the light (Eq. (14)) — adopted throughout Section II; the opposite quasi-static limit is addressed below.

### 2. Survival of the LHAASO TeV photons

The foam-modified spectrum of a broadband source splits into a coherent core and an incoherent pedestal,

$$S_{\rm obs}(\nu) = e^{-\sigma_\phi^2(\nu)}\, S_{\rm src}(\nu) + \int S_{\rm src}(\nu')\, \hat{R}(\nu-\nu';\nu')\, d\nu', \tag{21}$$

where the first term is the surviving carrier, suppressed per mode by $e^{-\sigma_\phi^2(\nu)}$, while the second is the flux removed from the core, $1-e^{-\sigma_\phi^2}$, redistributed by the pedestal kernel $\hat{R}$ of width $\sim 1/\tau_{\rm foam}$. Two assumptions govern whether the pedestal returns flux to the observing band.

*a. Assumption (i): dynamical foam.* The pedestal exists only if foam decorrelates, $C_\alpha(\tau) \to 0$; the quasi-static limit $C_\alpha \to 1$ yields $\mathcal{F}_{\rm foam} \to 1$ with no suppression to constrain (Section II B). For Planck-time-refresh foam ($\tau_{\rm foam} \sim \tau_P \equiv \ell_P/c \approx 5\times10^{-44}\,$s), this escape is closed even at TeV: the margin $\tau_c/\tau_{\rm foam} \gtrsim 10^{22}$ at 37 MeV shrinks toward higher energy as $\tau_c \propto 1/\nu$ (with $\tau_c = 1/(2\pi r_{\rm obs}\nu)$ at the constant fractional width $r_{\rm obs} = 0.10$), but remains $\gtrsim 10^{17}$ at 3 TeV — still enormous, so the foam is dynamical throughout. This energy-scaling argument does not address models with long-range temporal correlations (the temporal analogue of Ref. [10]; see Section II B), which give $C_\alpha \approx 1$ regardless of photon energy and remain a separate escape from all bounds in this paper.

*b. Assumption (ii): bandwidth ≪ pedestal.* The pedestal width is $\sim c/\ell_P \approx 10^{43}\,$Hz, whereas the entire electromagnetic output of a GRB (radio to TeV) spans $\lesssim 10^{27}\,$Hz, a ratio $\sim 10^{-16}$. For a Lorentzian pedestal of half-width $\Gamma = c/\ell_P \approx 1.9\times10^{43}\,$Hz, the in-band kernel fraction over the LHAASO acceptance (3–13 TeV, $\Delta\nu \approx 2.4\times10^{27}\,$Hz) is $f_{\rm band} = \Delta\nu/(\pi\Gamma) \approx 4\times10^{-17}$, so scattered flux returning to band is negligible. This evaluates the pedestal at $\tau_{\rm foam} = \tau_P$; more generally the in-band fraction is negligible only while the pedestal stays broader than the band, $1/\tau_{\rm foam} \gg \Delta\nu_{\rm band}$, i.e. $\tau_{\rm foam} \lesssim 1/\Delta\nu_{\rm band} \approx 4\times10^{-28}\,$s. This is far stronger than the dynamical-foam condition $\tau_{\rm foam} \ll \tau_c \sim 10^{-22}\,$s that the width bound requires: for $10^{-28}\,{\rm s} \lesssim \tau_{\rm foam} \lesssim 10^{-22}\,$s the foam is still dynamical (the line-width bound holds) but the pedestal narrows into the band and the TeV survival bound weakens. The TeV exclusion of $\alpha = 2/3$ therefore rests specifically on near-Planck-refresh foam, $\tau_{\rm foam} \lesssim 4\times10^{-28}\,$s.

With the in-band pedestal negligible, the observed flux reduces to the absolute suppression

$$S_{\rm obs}(\nu)\Big|_{\rm in\text{-}band} \;\approx\; e^{-\sigma_\phi^2(\nu)}\, S_{\rm src}(\nu). \tag{22}$$

This is an absolute attenuation, testable whenever the detected flux can be compared to an independently predicted intrinsic flux. Equation (22) holds while the surviving core exceeds the pedestal floor, $\sigma_\phi^2 \lesssim -\ln f_{\rm band} \approx 38$ (Section IV; distinct from the suppression strength $\sigma^2_{\phi,\rm max} \sim$ 1–10 that the data actually constrain, Eq. (23)); for stronger foam the in-band signal saturates at the floor $\sim f_{\rm band} N_0$, an escape foreclosed below by the transient, directional nature of the detection.

The LHAASO-KM2A detector registered 142 gamma rays above 3 TeV from GRB 221009A at $20.6\,\sigma$ significance, with events extending to $\sim$13 TeV [24, 25]. By Eq. (22) the observed in-band count is $N \approx e^{-\sigma_\phi^2(E_{\rm TeV})}\, N_0$, where $N_0$ is the in-band count in the absence of foam (after absorption by the extragalactic background light, EBL). Detection of $N$ photons implies

$$\sigma_\phi^2(E_{\rm TeV}) \;\approx\; \ln\frac{N_0}{N} \;\equiv\; \sigma^2_{\phi,\rm max}, \tag{23}$$

where $\sigma^2_{\phi,\rm max}$ is the number of $e$-foldings of suppression the source can sustain before the foam-attenuated count falls below the observed $N$. Writing $\sigma_\phi^2 = (2\pi D_C/\lambda_{\rm obs})^2(\ell_P/D_C)^{2\alpha}$ from Eqs. (1) and (4) ($A_\alpha = 1$, $\lambda_{\rm obs} = hc/E_{\rm TeV}$) and inverting gives

$$\alpha \;\gtrsim\; \frac{\ln\!\big(2\pi D_C/\lambda_{\rm obs}\big) - \tfrac{1}{2}\ln\sigma^2_{\phi,\rm max}}{\ln(D_C/\ell_P)} \;\equiv\; \alpha_{\rm TeV}. \tag{24}$$

The strength of this bound is best seen by reductio. At the holographic value $\alpha = 2/3$, Eqs. (1)–(4) at $E_{\rm TeV} = 3\,$TeV give $\sigma_\phi^2 \approx 7\times10^8$. To produce the observed 142 photons would therefore require an intrinsic in-band count $N_0 = N\, e^{\sigma_\phi^2} \approx 142\, e^{7\times10^8} \approx 10^{3\times10^8}$ — exceeding any astrophysical source by hundreds of millions of orders of magnitude. The exclusion is immune to any uncertainty in the intrinsic spectrum or EBL model.

The precise boundary depends on how many $e$-foldings the source can tolerate: allowing $\sigma^2_{\phi,\rm max} = 1$, 3, or 10 $e$-foldings gives $\alpha_{\rm TeV} = 0.740$, 0.736, or 0.732 respectively ($E_{\rm TeV} = 3\,$TeV, $\ln(2\pi D_C/\lambda_{\rm obs}) \approx 102.4$, $\ln(D_C/\ell_P) \approx 138.4$). At the highest detected energy $E_{\rm TeV} \approx 13\,$TeV the bound tightens to $\alpha \gtrsim 0.75$. The result is doubly logarithmic in

$N_0/N$: a decade in the tolerated suppression shifts $\alpha$ by less than 0.01 (at $z = 0.151$ the EBL optical depth at 3 TeV is only $\tau_{\rm EBL} \sim 0.3$).

The escape could instead lie in the pedestal kernel itself — a light-tailed $\hat{R}$ that, contrary to Assumption (ii), returns a non-negligible fraction of the redistributed flux to the observing band. The transient, directional nature of the detection forecloses this: the kernel acts on every source at all times, so appreciable band-refilling would deposit flux as a persistent sky-wide background — not as a $20.6\,\sigma$ excess from the direction of GRB 221009A, coincident with the burst. The 142 TeV photons are therefore genuine survivors of the source's beamed emission, and the suppression they constrain is real. The same reasoning bounds the kernel at the population level: a refilling kernel would reprocess the entire steady $\gamma$-ray source population into a diffuse glow far exceeding the observed background, an overproduction sharpest in the $\gamma$-ray band itself because $\sigma_\phi^2 \propto \nu^2$ concentrates the reprocessing at high energies. That detecting the suppression requires a once-in-$10^4$-year transient, rather than a uniformly glowing sky, is itself the statement that the in-band refill is negligible.

Dynamical foam redistributes intensity in frequency. The modulation $e^{i\omega\delta t(t)}$ has unit modulus, so $|E_{\rm obs}(t)|^2 = |E_{\rm src}(t)|^2$ pointwise and the integrated flux $\int S\,d\nu$ is conserved exactly; what the foam changes is its distribution in $\nu$, the flux removed from the core reappearing in the pedestal of width $\sim 1/\tau_{\rm foam}$. Because $C_\alpha$ is the autocorrelation of a stationary process, $\mathcal{F}_{\rm foam}$ is real and even in $\tau$ and the mean frequency of the full spectrum is unchanged: the core redshift $\nu_0^{\rm eff} < \nu_0$ is compensated by blue-weighted pedestal, so it records a redistribution between core and pedestal rather than a net transfer from the metric. Stationarity of $\delta t$ is time-translation invariance of the perturbation, and no net ensemble-mean energy transfer can follow from it. ~~Dynamical foam exchanges energy with the photon: the time-dependent metric acts as a reservoir. The per-mode modulation $e^{i\omega\delta t(t)}$ conserves photon number — $|e^{i\omega\delta t}| = 1$ on every mode, for any jitter distribution — but not energy, which it redistributes across the pedestal of width $\sim 1/\tau_{\rm foam}$; the predicted net carrier redshift $\nu_0^{\rm eff} < \nu_0$ is the direct signature of this transfer.~~ Because flux conservation and ~~Because number conservation and~~ the core-plus-pedestal split follow from unit modulus alone, they survive the extreme limit $\tau_{\rm foam} \to \tau_P$, where the pedestal width $1/\tau_{\rm foam}$ greatly exceeds the carrier $\omega_0$: that limit does not invalidate the decomposition, but drives $\sigma_\phi^2$ into the regime where the coherent core is exponentially suppressed.

This survival exclusion is consistent with the imaging-channel finding that GRB 221009A's localization, including the LHAASO TeV photons, admits $\alpha = 2/3$ [13]: imaging constrains the spatial foam correlations, which blur the wavefront but retain the photons, whereas the survival bound constrains the temporal (dynamical) correlations, which remove them from the band. The two coincide only for static foam, where neither produces a spectral effect.

These bounds should be set against the window $0.650 \le \alpha \le 0.735$ that Refs. [12, 13] obtain from imaging this same burst, a two-sided bracket with the holographic value at its midpoint rather than a one-sided limit. The bounds here come from a different channel and rest on different assumptions: a spectral rather than an angular observable, temporal rather than transverse foam correlations as just noted, and an intrinsic source spectrum in place of a known aperture. The question they answer is therefore whether a second route to $\alpha$ agrees with the first, and it does. The two pedestal-independent bounds lie below the window — $\alpha \gtrsim 0.63$ from the line width here, $\alpha \gtrsim 0.52$ from the CMB monopole of Section V B — so they are floors consistent with it and with the holographic value, and they exclude no part of it. The conditional TeV survival bound is the single exception, and it excludes the window only for near-Planck-refresh foam: if $\alpha$ lies in the imaging window, the survival of the LHAASO photons instead requires $\tau_{\rm foam} \gtrsim 4\times10^{-28}$ s, the pedestal having narrowed into the band so that the survival probe closes. An imaging detection at $\alpha \approx 2/3$ and the TeV survival of GRB 221009A are then jointly a statement about how fast the foam refreshes, which neither channel yields alone.

### 3. *Energy-dependent line width*

The line-width probe is complementary because it is relative: the absolute suppression $e^{-\sigma_\phi^2(\nu)}$ attenuates line peak and adjacent continuum identically, canceling in the fractional width $\sigma/E$. It therefore needs no absolute-flux model — foam is read off the line shape alone — and is pedestal-independent, immune to the redistribution physics contested above, surviving rejection of Assumption (ii), the only escape from the TeV bound. Its limitation is that, for a narrow line, the fractional width responds to foam only at second order in the jitter, so it constrains $\alpha$ only in the strong-foam regime and yields a significantly weaker, if independent, floor on $\alpha$.

*a. The intrinsic line as a Doppler invariant.* The MeV feature is understood as a Doppler-shifted line from the relativistic outflow — most plausibly the blueshifted 511 keV electron–positron annihilation line of pair-plasma clumps comoving with the jet [11, 21], though what follows needs only that it is a boosted line, not the specific transition. Its observed centroid is then the comoving line blueshifted by the bulk Doppler factor $\mathcal{D}$ of the emitting material, $E_{\rm line} = \mathcal{D}\,m_e c^2/(1+z)$, so the centroid measures $\mathcal{D}$ directly; the observed decline $E_{\rm line} \propto t^{-1}$ from $\approx 37$ to $\approx 6$ MeV tracks the falling jet Lorentz factor ($\Gamma \propto t^{-1}$, $\Gamma \approx 84$ at 37 MeV in the Zhang et al. modeling, viewed near $\theta \sim 1/\Gamma$ so that $\mathcal{D} \approx \Gamma$) as the outflow evolves [21]. Because a bulk boost multiplies every photon energy — line center and

line width alike — by the same $\mathcal{D}$, the fractional width $r_{\rm int} = \sigma'/E'$ is boost-invariant: it equals its comoving value at every epoch, independent of $\mathcal{D}$, and is therefore expected to stay constant as the burst evolves and the centroid sweeps downward. This is what is observed — $r_{\rm obs} = 0.10 \pm 0.01$ across the seven sub-intervals (Section V A 1) — so in the conventional picture $r_{\rm int}$ is a fixed, epoch-independent property of the source, set by the comoving line shape (the spread of clump Doppler factors and the pair-plasma temperature [21]) and carrying no foam. Quantum-gravity jitter, if present, imprints an additional, energy-dependent broadening on top of this constant baseline, computed next.

*b. Derivation of fractional-width broadening.* The width law below follows from the slowly-varying-envelope limit $\sigma_\Delta \ll \tau_G$, in which the foam factor multiplies the line cleanly; the 95% bound, however, lies at $\sigma_\Delta/\tau_G \approx 4.1$, so it extrapolates the core's width relation into the strong-foam regime — a tension quantified in the caveats below. Eq. (17) gives an observed core whose absolute width $\sigma_{\rm obs} = 1/[2\pi\sqrt{\tau_G^2 + \sigma_\Delta^2}]$ is narrower than the intrinsic $\sigma_\nu = 1/(2\pi\tau_G)$ (item (ii), Section III B); however, the line center shifts down to $\nu_0^{\rm eff} = \nu_0 \tau_G^2/(\tau_G^2 + \sigma_\Delta^2)$ (item (i)) by a larger relative amount. The fractional width is $r_{\rm obs} \equiv \sigma_{\rm obs}/\nu_0^{\rm eff}$, with $\nu_0^{\rm eff}$ the observed centroid. Combining gives

$$r_{\rm obs} = \frac{\sigma_{\rm obs}}{\nu_0^{\rm eff}} = \frac{1/(2\pi\sqrt{\tau_G^2 + \sigma_\Delta^2})}{\nu_0\, \tau_G^2/(\tau_G^2 + \sigma_\Delta^2)} = \frac{1}{2\pi\nu_0\tau_G}\,\frac{\sqrt{\tau_G^2 + \sigma_\Delta^2}}{\tau_G} = r_{\rm int}\sqrt{1 + \sigma_\Delta^2/\tau_G^2}, \tag{25}$$

so the fractional width broadens:

$$r_{\rm obs}^2(E) \;=\; r_{\rm int}^2\big[1 + \sigma_\Delta^2/\tau_G^2(E)\big] \;=\; r_{\rm int}^2 + r_{\rm foam}^2(E), \tag{26}$$

with $r_{\rm foam}^2(E) \equiv r_{\rm int}^2\,\sigma_\Delta^2/\tau_G^2(E)$. The line is a superposition over clumps with a spread of Doppler factors (Section V A 1), and Eq. (15) is nonlinear in $\tau_G$ through its prefactor and its carrier shift, so an intensity-weighted mean over clumps is not that closed form at a mean $\tau_G$; it is applied to the line as a whole, with $r_{\rm int}$ the fractional width of the composite, which holds to second order in $\sigma_\Delta^2/\tau_G^2$, the order of the closed form's other corrections.

In the dynamical limit $\sigma_\Delta = \sqrt{2}\,\sigma_\ell/c$ saturates, and with $\tau_G(E) = \lambda_{\rm obs}(E)/(2\pi c\, r_{\rm int})$ the foam excess becomes

$$r_{\rm foam}(E) \;=\; \sqrt{2}\, r_{\rm int}^2\, \sigma_\phi(E) \;=\; r_{\rm int}^2 \cdot \frac{2\sqrt{2}\,\pi\,\sigma_\ell}{\lambda_{\rm obs}(E)} \;\propto\; E, \tag{27}$$

predicting chromatic broadening that grows linearly with photon energy. The $r_{\rm int}^2$ prefactor makes the fractional width a second-order foam probe: since $\sigma_\Delta^2/\tau_G^2 = 2r_{\rm int}^2\sigma_\phi^2$, an order-unity broadening of a narrow line ($r_{\rm int} \approx 0.1$) requires $\sigma_\phi^2 \gtrsim 40$. At the holographic value $\alpha = 2/3$ this threshold is reached at $E_{\rm obs} \approx 720\,{\rm MeV}$ (where $\sigma_\phi^2 = 40$); a confirmed narrow line at or above this energy in a $z \sim 0.15$ GRB would exclude $\alpha = 2/3$ through the width alone, independent of the pedestal kernel. Like the width bound itself, this test operates in the strong-foam regime where the coherent core is exponentially suppressed; it asks whether the line survives as a coherent feature at all.

*c. Fit and bound.* A $\chi^2$ fit of the seven measurements to the two-parameter model

$$r_{\rm obs}^2(E) \;=\; r_{\rm int}^2 + r_{\rm foam,0}^2\,(E/E_0)^2, \qquad E_0 = 37\,{\rm MeV}, \tag{28}$$

constrains the foam amplitude. The best fit lies at the boundary $r_{\rm foam,0} = 0$ (with $r_{\rm int} = 0.111$, $\chi^2_{\rm min} = 5.3/6\,{\rm dof}$), confirming the constant-$r_{\rm obs}$ hypothesis. Profiling over $r_{\rm int}$ gives the 95% CL upper limit $r_{\rm foam,0}^{\rm max} = 0.27$ at $\Delta\chi^2 = 2.71$ (Fig. 1; boundary-corrected Chernoff threshold [26]: $\frac{1}{2}\delta_0 + \frac{1}{2}\chi_1^2$ for zero-bounded parameters; the precise threshold barely affects $\alpha$, the logarithmic lever arm dominating). Converting via Eq. (27) and $\sigma_\ell^2 = D_C^{2(1-\alpha)}\ell_P^{2\alpha}$ (Eq. (1), $A_\alpha = 1$) gives

$$\alpha \;\gtrsim\; \frac{\ln(D_C/\lambda_{\rm obs}) - \ln r_{\rm foam,0}^{\rm max} + \ln(2\sqrt{2}\,\pi\, r_{\rm int}^2)}{\ln(D_C/\ell_P)} \;=\; \frac{89.3 + 1.3 - 3.3}{138.4} \;\approx\; 0.63\,. \tag{29}$$

Here $r_{\rm int} = 0.066$ is the width profiled jointly with $r_{\rm foam,0}^{\rm max}$ on the 95% boundary (not the global best fit 0.111), giving $|\ln(2\sqrt{2}\,\pi\, r_{\rm int}^2)| \approx 3.3$. Like the TeV bound, the result is doubly logarithmic and so dominated by the kinematic lever arm $\ln(D_C/\ell_P) = 138$: a tenfold change in $r_{\rm foam,0}^{\rm max}$ moves $\alpha$ by only $\ln 10/\ln(D_C/\ell_P) \approx 0.017$, so $\alpha \gtrsim 0.63$ is fixed by the geometry, not by the precision of the seven measurements.

*d. Validity and caveats.* At the 95% boundary $\sigma_\Delta/\tau_G(E_0) \approx 4.1$, i.e. $\sigma_\phi^2(37\,{\rm MeV}) \sim 10^3$, the bound lies deep in the strong-foam regime, and three caveats attach to it. First, the fractional-width law Eq. (26) is a property of the coherent core, whose weight $e^{-\sigma_\phi^2}$ is exponentially small there; the bound thus extrapolates the core's width relation into a regime where the line flux has largely been redistributed into the pedestal, and is best read as an

TABLE I. The seven Zhang et al. [21] MeV-line sub-intervals used in the line-width fit: post-trigger interval, line energy $E$, observed fractional width $r_{\rm obs} = \sigma_{\rm obs}/E$, and its uncertainty $\sigma_r$. The weighted mean is $r_{\rm obs} = 0.111 \pm 0.014$; the values are consistent with a constant ($\chi^2/\nu = 5.3/6 = 0.88$).

| interval [s] | $E$ [MeV] | $r_{\rm obs}$ | $\sigma_r$ |
|---|---|---|---|
| 246–256 | 37.2 | 0.191 | 0.088 |
| 275–280 | 17.8 | 0.093 | 0.041 |
| 285–290 | 15.0 | 0.129 | 0.046 |
| 290–295 | 12.1 | 0.120 | 0.030 |
| 295–300 | 12.5 | 0.067 | 0.031 |
| 300–310 | 8.9 | 0.196 | 0.061 |
| 310–320 | 9.7 | 0.115 | 0.030 |

order-of-magnitude exclusion anchored by the logarithmic lever arm rather than a sharp threshold. Second, the same Eq. (15) predicts a centroid depression $\nu_0^{\rm eff}/\nu_0 = \tau_G^2/(\tau_G^2 + \sigma_\Delta^2) \approx 0.06$ at 37 MeV at this boundary, absent from the smooth $E_{\rm line} \propto t^{-1}$ trend in the data; the width-only fit treats observed centroids as intrinsic because a foam redshift is degenerate with the kinematic $E_{\rm line} \propto t^{-1}$ evolution the data already exhibit. The centroid is the first-order, in principle stronger, foam probe, so the second-order width bound is conservative; a joint centroid-plus-width analysis is left to future work. Third, the bound rests on the seven sub-interval widths of the single GRB 221009A feature, whose most constraining energy (37 MeV) lies in the Fermi MeV sensitivity gap (Section VI), and the constant-$r_{\rm int}$ fit cannot distinguish a genuine null from small epoch-to-epoch variation of the intrinsic width. With these caveats, the width-only $\alpha \gtrsim 0.63$ stands as the pedestal-independent floor beneath the TeV bound.

## B. CMB

The CMB blackbody is itself a broad, asymmetric spectral profile, so foam acts on it exactly as on any line: the coherent core is suppressed by $e^{-\sigma_\phi^2(\nu)}$ and the removed flux reappears as a broad pedestal whose shape depends on the short-time behavior of $C_\alpha(\tau)$. Because $\sigma_\phi^2 \propto \nu^2$ (Eq. (4)), the foam deficit is concentrated at high frequencies — a $\nu^2$-weighted suppression distinct from the standard chemical-potential ($\mu$) and Compton-$y$ distortions. For Planck-scale foam the pedestal is so broad that it lies almost entirely outside the observing band (Section V B 2), leaving an essentially pure deficit in the FIRAS window.

The Far-InfraRed Absolute Spectrophotometer (FIRAS) aboard the Cosmic Background Explorer (COBE) measured the absolute spectrum of the CMB monopole over 60–600 GHz by direct comparison against an internal cryogenic blackbody calibrator — the absolute-photometry feature that distinguishes it from relative-photometry experiments and makes it well suited to constraining spectral shape distortions. The calibrated monopole agrees with a pure blackbody to within $\sim 50$ ppm (a few $\times 10^{-5}$ fractional) across the band, placing the still-standard limits $|\mu| \lesssim 9 \times 10^{-5}$ and $|y| \lesssim 1.5 \times 10^{-5}$ [27]. These limits and the 43-channel calibrated spectrum form the empirical basis for the foam constraint derived below.

### 1. *Foam phase variance at CMB frequencies*

CMB photons travel from the last-scattering surface at comoving distance $D_{\rm C} = 13.87$ Gpc (Planck 2018 flat-ΛCDM cosmology [23]; $H_0 = 67.36\,{\rm km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315$). The foam phase variance (Eq. (4)) is set by the effective foam amplitude $A_{\rm eff} = A_\alpha D_{\rm C}^{2(1-\alpha)} \ell_P^{2\alpha}$ (the lumped amplitude entering the FIRAS fit, Section V B 3):

$$\sigma_\phi^2(\nu) = \left(\frac{2\pi\nu}{c}\right)^2 A_{\rm eff}, \tag{30}$$

with $A_\alpha = 1$.

At $\nu = 150$ GHz this gives

$$\sigma_\phi^2(150\,{\rm GHz}) \approx \begin{cases} 7 \times 10^{-2} & \alpha = 1/2 \text{ (random walk)}, \\ 2 \times 10^{-22} & \alpha = 2/3 \text{ (holographic)}. \end{cases} \tag{31}$$

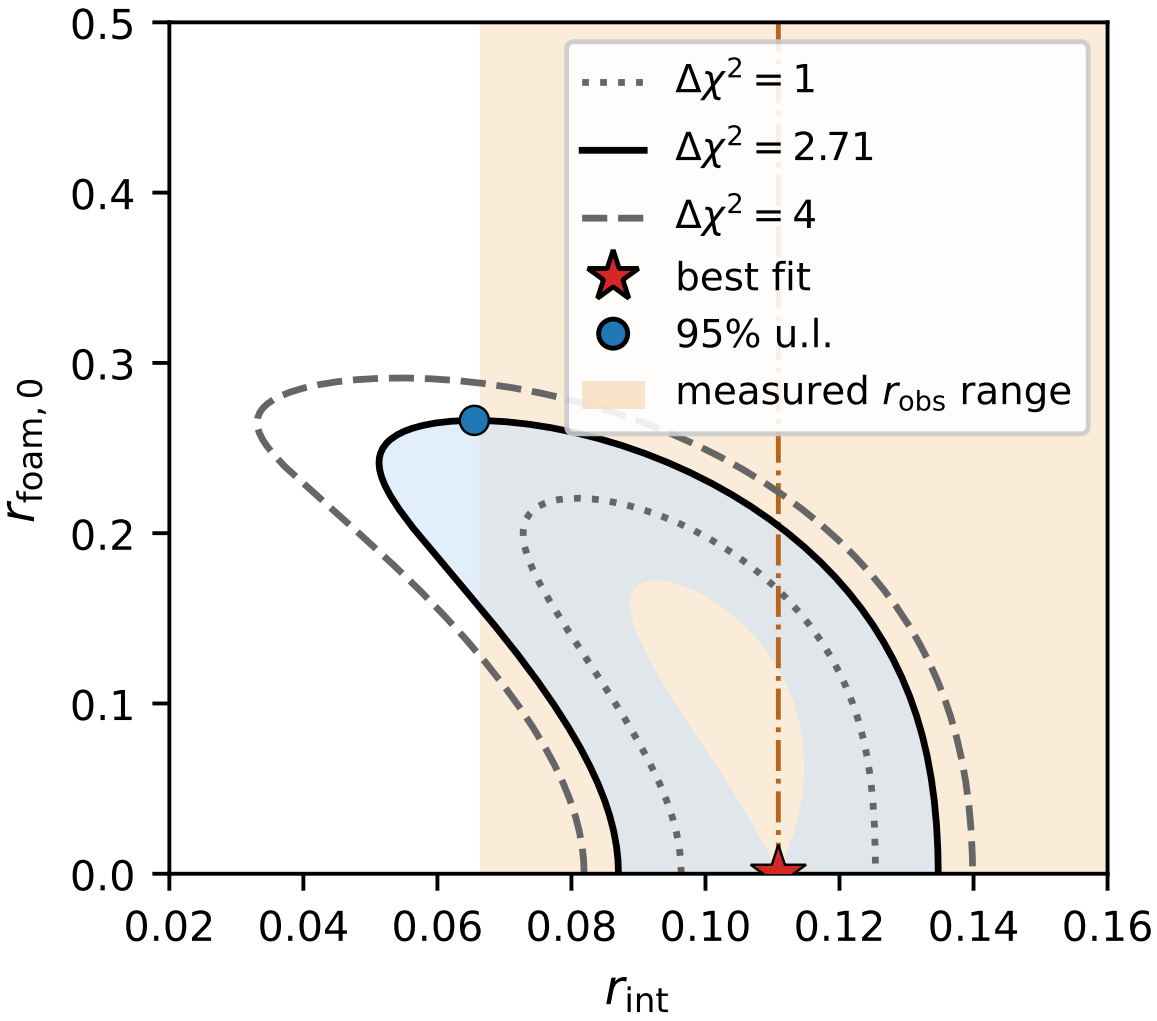



FIG. 1. $\Delta\chi^2$ in the $(r_{\rm int}, r_{\rm foam,0})$ plane for the line-width fit (contours at 1, 2.71, 4). The best fit lies at the boundary $r_{\rm foam,0} = 0$, $r_{\rm int} = 0.111$ (star); the one-sided 95% upper limit $r^{\rm max}_{\rm foam,0} = 0.27$ is reached on the $\Delta\chi^2 = 2.71$ contour at the profiled $r_{\rm int} = 0.066$ (circle). The shaded band marks the measured range of $r_{\rm obs}$ (Table I); the limit trades a narrower intrinsic line against larger foam amplitude.

For $\alpha = 1/2$ the variance reaches order unity at the top of the FIRAS band ($\sigma^2_\phi \approx 1.2$ at 639 GHz), and the linearised template holds for $\alpha \gtrsim 0.50$, where $\sigma^2_\phi(\nu) \lesssim 1$ through the band (Section V B 3).

### 2. *Foam-modified CMB spectrum*

Given a foam model, the natural analysis is of the whole modified spectrum: compute $g^{(1)}_{\rm src}(\tau)$ as the normalized Fourier transform of $B_\nu(T_0)$ (Eq. (16)), apply the jitter convolution (11), and inverse-transform to $S_{\rm obs}(\nu)$. We instead separate it into a coherent core and a pedestal, because the split isolates a model-independent piece — the suppression $e^{-\sigma^2_\phi(\nu)}$, exact for any $C_\alpha$ (Section IV) — from the model-dependent pedestal kernel $\hat{R}(\nu - \nu'; \nu')$, a decomposition that yields general physical insight the combined spectrum obscures. A worked example with exponential $C_\alpha(\tau)$ below identifies the physical no-pedestal case used in Section V B 3.

Decomposing the blackbody as an incoherent superposition of narrowband modes $S_{\rm src}(\nu') = B_{\nu'}(T_0)$, each obeys the same achromatic jitter with per-mode foam factor

$$\mathcal{F}_{\rm foam}(\tau; \nu') = e^{-\sigma^2_\phi(\nu')[1 - C_\alpha(\tau)]}, \quad \sigma^2_\phi(\nu') = (2\pi\nu'/c)^2 \sigma^2_\ell(D), \tag{32}$$

with $\sigma_\ell^2(D) = A_{\rm eff}$ (Section V B 1). By the core-plus-pedestal split of Section IV, in the dynamical limit each mode's long-lag plateau $e^{-\sigma_\phi^2(\nu')}$ Fourier-transforms to a coherent core at $\nu'$ and its short-lag residual to a broad pedestal; superposing over the Planck modes (intensity-weighted Eq. (16)) gives

$$S_{\rm obs}(\nu) = e^{-\sigma_\phi^2(\nu)}\, B_\nu(T_0) + \int_0^\infty B_{\nu'}(T_0)\, \hat{R}(\nu-\nu';\nu')\, d\nu', \tag{33}$$

with $\hat{R}$ the Fourier transform of the residual, carrying the full $C_\alpha$-dependence. By the unitarity of the per-mode modulation $e^{i\omega\delta t(t)}$, the pedestal conserves the integrated flux, $\int [S_{\rm obs}(\nu) - B_\nu(T_0)]\, d\nu = 0$, the displaced flux emerging at $\nu \sim 1/\tau_{\rm foam}$ ~~conserves photon number, $\int [S_{\rm obs}(\nu) - B_\nu(T_0)]\, d\nu/(h\nu) = 0$~~.

To make the pedestal kernel concrete, we evaluate $\hat{R}$ for the minimal model $C_\alpha(\tau) = e^{-|\tau|/\tau_{\rm foam}}$ (a Markovian metric fluctuation). Substituting into Eq. (7) gives

$$\sigma_\Delta^2(\tau) = \frac{2\sigma_\ell^2(D)}{c^2}\left[1 - e^{-|\tau|/\tau_{\rm foam}}\right]. \tag{34}$$

The Fisher analysis below operates in the linearised regime $\sigma_\phi^2(\nu') \ll 1$. Expanding $\mathcal{F}_{\rm foam}(\tau;\nu')$ (Eq. (32)) to first order and subtracting the long-lag asymptote $e^{-\sigma_\phi^2(\nu')}$ leaves the short-$\tau$ residual

$$\mathcal{F}_{\rm foam}(\tau;\nu') - e^{-\sigma_\phi^2(\nu')} \;\approx\; \sigma_\phi^2(\nu')\, e^{-|\tau|/\tau_{\rm foam}}, \tag{35}$$

whose Fourier transform is a Lorentzian centered on the source frequency,

$$\hat{R}(\nu-\nu';\nu') \;=\; \sigma_\phi^2(\nu')\, \frac{2\tau_{\rm foam}}{1 + [2\pi(\nu-\nu')\tau_{\rm foam}]^2}, \tag{36}$$

of full width at half maximum $1/(\pi\tau_{\rm foam})$. For Planck-scale foam with $\tau_{\rm foam} \sim \ell_P/c \approx 5\times10^{-44}\,$s, the Lorentzian width ($\sim 10^{43}\,$Hz) lies thirty-one orders of magnitude beyond the FIRAS upper edge ($\nu_{\rm max} = 6\times10^{11}\,$Hz), so the in-band flux fraction is $\sim 10^{-31}$. The physical pedestal thus deposits no detectable in-band power for any plausible $\tau_{\rm foam}$. The constraint of Section V B 3 consequently adopts the no-pedestal limit as the primary template.

### 3. *Constraint from FIRAS*

We use $I_\nu \equiv S_{\rm obs}(\nu)$ to denote the observed CMB intensity including the foam signal. Working in the linearised regime $\sigma_\phi^2(\nu) \ll 1$, we treat four spectral parameters $\boldsymbol{\theta} = (A_{\rm eff}, T_0, \mu, y)$, where $A_{\rm eff} = A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}$ is the cumulative comoving foam variance at the CMB distance (Section V B 1), so that $\sigma_\phi^2(\nu) = (2\pi\nu/c)^2 A_{\rm eff}$. Note that $\alpha$ is not fitted directly; only the lumped amplitude $A_{\rm eff}$ enters the fit, and bounds on $\alpha$ are obtained *a posteriori* by inverting the Ng–Perlman relation at fixed $A_\alpha = 1$.

For physical foam the pedestal contributes no in-band power (Section V B 2), so the only in-band foam signature is the pure deficit — the no-pedestal limit of Eq. (33) —

$$\frac{\partial I_\nu}{\partial A_{\rm eff}} = -\left(\frac{2\pi\nu}{c}\right)^2 B_\nu(T_0), \tag{37}$$

a $\nu^2$-weighted (blue-tilted) Planck spectrum. The chemical-potential and Compton-$y$ distortion templates are

$$\frac{\partial I_\nu}{\partial \mu} = -B_\nu(T_0)\, \frac{x e^x}{(e^x-1)^2}, \tag{38}$$

$$\frac{\partial I_\nu}{\partial y} = \;\; B_\nu(T_0)\, \frac{x e^x}{e^x-1}\left[\frac{x(e^x+1)}{e^x-1} - 4\right], \tag{39}$$

with $x = h\nu/(k_B T_0)$, encoding respectively early-universe energy injection (Silk damping, decaying particles) and late-time electron heating (reionization, cluster gas).

We constrain $A_{\rm eff}$ by fitting these templates — together with the FIRAS Galactic-foreground template — to the public COBE/FIRAS monopole residual spectrum and its calibration covariance [27]. Raising $T_0$ brightens the Planck curve broadband; the foam deficit suppresses it with an extra $\nu^2$ tilt. Over the CMB-bright band (peak through the onset of the Wien fall-off), the two differ only by this tilt, making the foam template nearly indistinguishable from a

temperature shift ($\rho_{A_{\rm eff}T_0} \simeq 1$ in the fit) and thus strongly collinear with the $y$ and $\mu$ templates. Using the full $43\times43$ apodization covariance, the generalized least-squares fit over $(T_0, \mu, y, A_{\rm eff})$ and the foreground amplitude recovers $\mu$ and $y$ consistent with zero (foam-marginalized 95% limits $|\mu| < 1.7\times10^{-4}$, $|y| < 1.0\times10^{-5}$; $\chi^2/\nu = 1.27$ — the $\mu$ limit looser than the foam-free Fixsen value $|\mu| \lesssim 9\times10^{-5}$ because marginalizing over the foam amplitude inflates $\sigma(\mu)$) and a foam amplitude consistent with zero,

$$A_{\rm eff} = (-2.4 \pm 5.3)\times10^{-11}\,{\rm m}^2. \tag{40}$$

Inverting $A_{\rm eff} = A_\alpha D_C^{2(1-\alpha)}\ell_P^{2\alpha}$ at $A_\alpha = 1$, the one-sided 95% upper limit (measured from zero, as required by the physical boundary $A_{\rm eff} \geq 0$) $A_{\rm eff} < 1.645\,\sigma(A_{\rm eff}) = 8.7\times10^{-11}\,{\rm m}^2$ gives the lower bound $\alpha \gtrsim 0.52$ (collected with the forecasts below in Table II). This bound is degeneracy-limited: $\alpha \gtrsim 0.52$ rests only on the residual spectral-tilt information that distinguishes the $\nu^2$-weighted deficit from a temperature shift, so marginalizing over $(T_0, \mu, y)$ leaves it weak. The only direction that could break the foam–$T_0$ degeneracy is deep into the Wien/sub-mm regime, where the $\nu^2$-tilt separates from a temperature shift; but there, beyond the FIRAS band, galactic dust and the cosmic infrared background dominate, making the monopole impractical. The fit uses the linearised template $\Delta I_\nu \approx \sigma_\phi^2(\nu)$ [template] (valid for $\sigma_\phi^2 \ll 1$), which holds across the FIRAS band at the headline threshold: $\sigma_\phi^2$ exceeds unity only for stronger, excluded foam ($\alpha \lesssim 0.50$; Section V B 1). Replacing the linearised deficit $-\sigma_\phi^2 B_\nu$ by the exact form $-(1-e^{-\sigma_\phi^2})B_\nu$ (valid at all $\sigma_\phi^2$, Section IV) shifts the integrated in-band amplitude by $\approx 13\%$ at $\alpha = 0.50$ but by $< 1\%$ at $\alpha \gtrsim 0.52$ ($\Delta\alpha \sim 3\times10^{-5}$), so the bound is unaffected; in that strong-foam regime nonlinearity may even help break the foam–$\mu$ degeneracy.

The foam–$T_0$ degeneracy has two consequences for the CMB temperature itself. First, admitting foam as a free parameter inflates the marginalized temperature uncertainty: the FIRAS spectral-shape determination of $T_0$ degrades from $1.2\times10^{-5}\,{\rm K}$ to $7.7\times10^{-4}\,{\rm K}$ (a factor $\sim 64$; $\sigma(y)$ and $\sigma(\mu)$ inflate by $\sim 17$ and $\sim 3$), comparable to the full COBE/FIRAS calibration uncertainty $5.7\times10^{-4}\,{\rm K}$ [28]. A foam-agnostic monopole analysis thus roughly doubles the temperature error budget, and the quoted precision of $T_0 = 2.72548\,{\rm K}$ implicitly assumes no foam. Second, foam present but omitted from the fit shifts the central value by $\Delta T_0 = b_{T_0} A_{\rm eff}$ with $b_{T_0} \approx -1.5\times10^7\,{\rm K\,m^{-2}}$; for $A_\alpha = 1$ this is only $\approx 3\times10^{-22}\,{\rm K}$ at the holographic $\alpha = 2/3$ and vanishes as $\alpha \to 1$ (where the comoving foam variance ceases to grow with distance), reaching $\sigma(T_0)$ only at $\alpha \approx 0.52$ where the foam is already a many-$\sigma$ signal. Both effects therefore matter only for foam near FIRAS detectability; the recovered $T_0$ is immune to any foam consistent with the GRB bound $\alpha \gtrsim 0.63$.

#### 4. *Fisher forecasts for future spectrometers*

To forecast future absolute spectrometers we propagate the no-pedestal templates through a Fisher matrix with flat per-channel (white) noise $\sigma_\nu = 0.01\,{\rm MJy/sr}$ calibrated to FIRAS (the full $43\times43$ apodization covariance enters the data fit of Section V B 3). The forecast gives $\sigma(A_{\rm eff}) = 2.2\times10^{-11}\,{\rm m}^2$ and $\alpha \gtrsim 0.52$ (reproducing the data fit); the conditional value $\sigma(A_{\rm eff})|_{\mu,y\,\rm fixed} = 8.2\times10^{-13}\,{\rm m}^2$ shows that marginalizing over distortions costs a factor $\sim 27$ via the foam–$T_0$ collinearity. The fiducial degeneracy structure is captured by the correlation matrix for ($A_{\rm eff}$, $T_0$, $\mu$, $y$),

$$\rho_{ij}^{\rm fid} = \begin{pmatrix} 1.000 & \mathbf{0.9997} & 0.928 & 0.999 \\ \mathbf{0.9997} & 1.000 & 0.934 & 0.998 \\ 0.928 & 0.934 & 1.000 & 0.915 \\ 0.999 & 0.998 & 0.915 & 1.000 \end{pmatrix}, \tag{41}$$

the foam deficit is almost perfectly aligned with a temperature shift ($\rho_{A_{\rm eff}T_0} = 0.9997$, bold) and strongly collinear with the $y$ and $\mu$ distortions (0.999 and 0.928). Defining the marginal detection SNR as $A_{\rm eff}/\sigma(A_{\rm eff})$, so that SNR = 1.645 marks the one-sided 95% CL threshold, Figure 2 plots it versus $\alpha$, and Table II collects the thresholds.

The no-pedestal limit is also the conservative one. The pure $\nu^2$-deficit is already near-perfectly aligned with a temperature shift ($\rho_{A_{\rm eff}T_0} = 0.9997$); any in-band pedestal redistributes power back into the band, moving the foam template off that alignment, so it can only tighten the bound. A $\mu$-shaped kernel, for instance, trades the near-perfect $T_0$ degeneracy for a weaker foam–$\mu$ one (0.976), nudging the threshold to $\alpha \gtrsim 0.53$. This is moot for physical foam, whose pedestal is negligible in-band for any $C_\alpha$ with $\tau_{\rm foam} \ll 1/\nu_{\rm max}$ (Section V B 2); an external $\mu$ prior likewise barely helps, since the limiting degeneracy is with $T_0$, not $\mu$.

The proposed PIXIE satellite [30] — a fully cryogenic ($T_{\rm sys} \approx 2.7\,{\rm K}$) nulling polarizing Fourier transform spectrometer spanning 30 GHz to 6 THz — is projected to surpass FIRAS sensitivity by roughly three orders of magnitude, reaching $|\mu| \lesssim 10^{-8}$ and $|y| \lesssim 10^{-9}$ and entering the regime of guaranteed ΛCDM spectral signals from Silk damping

TABLE II. Marginal foam-detection thresholds (one-sided 95% CL). Lower bounds on $\alpha$ ($A_\alpha = 1$): the foam signal $\sigma_\ell^2 \propto (\ell_P/D_C)^{2\alpha}$ decreases with $\alpha$, so non-detection excludes small $\alpha$. FIRAS: data fit (Section V B 3). PIXIE, SuperPIXIE, Voyage 2050: Fisher forecast scaled to published $\mu$-sensitivities [29]. The three forecast entries (0.545, 0.549, 0.554) round to 0.54–0.55.

| | lower bound |
|---|---|
| FIRAS | $\alpha \gtrsim 0.52$ |
| PIXIE | $\alpha \gtrsim 0.54$ |
| SuperPIXIE | $\alpha \gtrsim 0.55$ |
| Voyage 2050 | $\alpha \gtrsim 0.55$ |

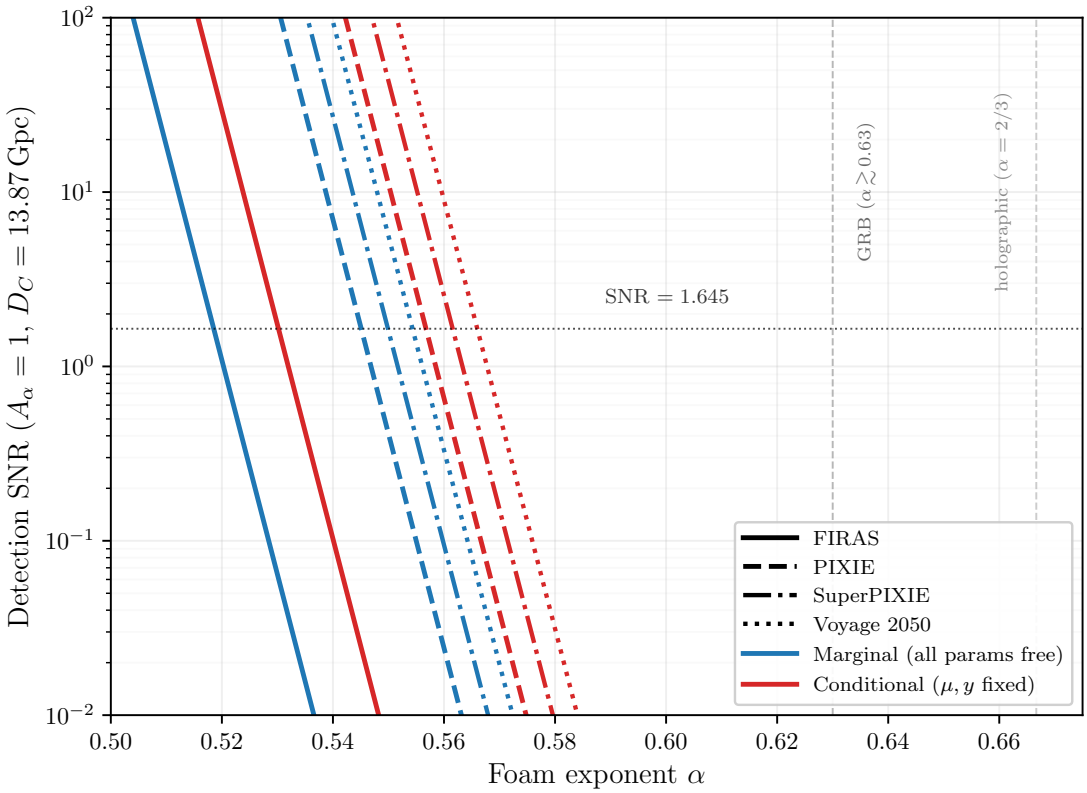



FIG. 2. Detection SNR for the foam amplitude $A_{\rm eff}$ versus exponent $\alpha$ ($A_\alpha = 1$, $D_C = 13.87$ Gpc). Blue: marginal SNR (all four parameters free). Red: conditional SNR ($\mu, y$ fixed), $\sim 27\times$ higher. Line styles: solid FIRAS, dashed PIXIE, dash-dotted SuperPIXIE, dotted Voyage 2050 (forecast $\mu$-sensitivities scaled to FIRAS; [29]). The marginal lower bounds cluster at $\alpha \approx 0.52$–$0.55$. Horizontal dotted line: SNR = 1.645 (one-sided 95% CL). Vertical dashed lines: the spectroscopic lower bound from GRB 221009A ($\alpha \gtrsim 0.63$) and the holographic prediction $\alpha = 2/3$.

and reionization. Propagating this gain through the Fisher-matrix scaling derived above, the marginal foam detection threshold shifts from $\alpha \gtrsim 0.52$ (FIRAS) to approximately $\alpha \gtrsim 0.54$ (PIXIE; Table II), narrowing the gap with the spectroscopic lower bound $\alpha \gtrsim 0.63$ from GRB 221009A (Fig. 2). The foam–$T_0$ degeneracy persists on the linearized template. But PIXIE's extension to 6 THz enters the nonlinear regime $\sigma_\phi^2 \gg 1$ for $\alpha \lesssim 0.52$ (higher than the FIRAS crossing $\alpha \lesssim 0.50$ because the wider band pushes the $\sigma_\phi^2 = 1$ boundary upward; Section V B 3), where the foam deficit diverges markedly from the blackbody and $\mu$-distortion shapes, possibly reducing the degeneracy. The $\alpha \gtrsim 0.54$ projection may therefore understate the reach: if the nonlinear template effectively breaks this degeneracy at PIXIE sensitivity, the true threshold could be significantly more stringent.

Extending the same Fisher analysis to the 10 GHz–3 THz bands of the SuperPIXIE and Voyage 2050 concepts [29] (forecast distortion sensitivities $\sigma(\mu) \approx 7.7 \times 10^{-9}$ and $2.2 \times 10^{-9}$) shifts the marginal threshold only to $\alpha \gtrsim 0.55$ (Table II), barely beyond the PIXIE projection. The reason is structural rather than instrumental: the foam deficit $-(2\pi\nu/c)^2 B_\nu$ and the temperature derivative are nearly proportional across the band, so the foam–$T_0$ degeneracy is

set by the templates, not the noise weighting. Repeating the Fisher analysis with the full $43 \times 43$ FIRAS apodization covariance rather than the white-noise forecast gives $\rho_{A_{\rm eff}T_0} = 0.9999$ (and a marginal-to-conditional inflation $\approx 25$) — the degeneracy is thus marginally more severe under realistic correlated noise, so the white-noise estimate is conservative. The wider band does break the foam–$\mu$ degeneracy that an in-band $\mu$-shaped pedestal would exploit (from $\rho_{A_{\rm eff}\mu} = 0.98$ at FIRAS to 0.66 over 10 GHz–3 THz), so future missions are markedly more robust against in-band pedestal contamination, but, within the linearized analysis, the physical threshold does not move (the nonlinear, strong-foam exception noted above, $\alpha \lesssim 0.52$, aside). The progression saturates logarithmically: each decade of $\sigma(\mu)$ buys only $\Delta\alpha = \ln 10/[2\ln(D_C/\ell_P)] \approx 0.008$, so the $\sim$ 3-decade FIRAS→PIXIE gain gives the $\approx 0.03$ step while PIXIE→Voyage 2050 (under a decade) adds $\approx 0.009$ (the forecast entries 0.545, 0.549, 0.554 round to 0.54–0.55). Even a $10^4$-fold sensitivity gain over FIRAS converts to $\Delta\alpha \lesssim 0.04$, a firm ceiling: as a broadband probe based on absolute calibrated photometry (no source-profile assumption, a different systematic-error budget from spectroscopy), spectral-distortion photometry yields $\alpha \gtrsim 0.52$ from FIRAS and at most $\alpha \gtrsim 0.55$ next-generation, the bound limited by the foam–$T_0$ degeneracy rather than by sensitivity or bandwidth. In short, no absolute-photometry CMB spectrometer reaches the holographic $\alpha = 2/3$: the limitation is the foam–$T_0$ degeneracy, not instrumental sensitivity, so the high-frequency GRB channel is essential to test holographic foam.

## VI. CONCLUSIONS

Spacetime foam imprints an achromatic, stochastic arrival-time jitter that modifies the first-order temporal coherence $g^{(1)}(\tau)$, and hence the observed spectrum, providing a spectroscopic probe complementary to the imaging (PSF, Strehl) channel. In the dynamical-foam limit the spectrum splits into a coherent line core suppressed by $e^{-\sigma_\phi^2}$ and an incoherent pedestal, yielding a shape-distortion bound on the foam exponent~~Ng–Perlman exponent~~ $\alpha$. The MeV emission lines of GRB 221009A give $\alpha \gtrsim 0.63$ (one-sided 95% CL), while a fit to the COBE/FIRAS monopole gives $\alpha \gtrsim 0.52$ (95% CL); next-generation spectrometers (PIXIE, SuperPIXIE, Voyage 2050) reach $\alpha \gtrsim 0.54$–$0.55$, the bound saturating logarithmically because the foam deficit is nearly degenerate with a temperature shift. All three are dominated by this kinematic lever arm rather than by the data: improved photon counts, line-width precision, or FIRAS-class sensitivity would shift $\alpha$ by only $\sim 0.01$–$0.02$. The pedestal-independent line-width and CMB bounds do not reach the holographic value $\alpha = 2/3$; only the conditional TeV survival bound ($\alpha_{\rm TeV} \gtrsim 0.74$, for near-Planck-refresh foam, $\tau_{\rm foam} \lesssim 4 \times 10^{-28}$ s; Assumption (ii)) would exclude it. The first two are floors below the imaging window $0.650 \leq \alpha \leq 0.735$ reported for this source [12, 13] and exclude no part of it: the spectroscopic channel is an independent check on that result rather than a competitor to it. Combined with that window, the third bounds $\tau_{\rm foam}$ from below rather than excluding $\alpha = 2/3$ (Section V A 2). Because the two channels probe the same exponent through different sources, observables, and systematics, they form an independent cross-check rather than a joint bracket. They are separated by more than ten orders of magnitude in frequency ($\sim 10^{11}$ Hz for the CMB versus $10^{22}$–$10^{27}$ Hz for the GRB), and since the foam phase variance grows as $\sigma_\phi^2 \propto \nu^2$, this $\nu^2$ scaling makes the high-frequency GRB channel intrinsically more sensitive, hence the more constraining. This sensitivity advantage is partly offset by a firmer baseline on the CMB side: its intrinsic spectrum is a near-perfect blackbody, whereas the GRB bound depends on a less certain model of the source's intrinsic emission.

All single-source spectroscopic bounds derived in this paper assume the dynamical-foam limit $\tau_{\rm foam} \ll \tau_c$ (Section III B), in which the foam path-length jitter decorrelates many times within each coherence time. If foam fluctuations are instead slow ($\tau_{\rm foam} \gg \tau_c$), the accumulated phase is essentially frozen during each coherence time and the foam acts as a deterministic, frequency-independent phase screen — no spectral broadening or peak reduction occurs within a single observation. The CMB constraint (Section V B) is derived in the same dynamical limit ($C_\alpha \to 0$; Section V B 2): in the quasi-static limit the per-mode foam suppression $\mathcal{F}_{\rm foam}(\tau) \to 1$ (Eq. (13)), leaving the monopole unmodified. In both channels the quasi-static foam thus produces no shape distortion. A separate escape operates even for nominally dynamical foam: long-range temporal correlations — the temporal analogue of the transverse spatial correlations of Ref. [10] (Section II B) — would hold $C_\alpha(\tau) \approx 1$ across observational lags regardless of photon energy, again nulling the shape-distortion signal. A third condition is external to the foam model: the bounds are based on Eq. (4), which does not account for the diffractive averaging of foam structure transverse to the ray (Section II A) — as distinct from foam structure across the collecting aperture, which is accounted for and is harmless when the whole image is recorded (Appendix A). The proper procedure is to compute $C(\mathbf{r}, \tau)$ for candidate foam models, as done in Lee *et al.* [10].

The dynamical bound $\alpha \gtrsim 0.63$ (Eq. (29)) is a weak, lever-arm-dominated floor; tightening it is limited primarily by an instrumental gap. The spectral region between the upper edge of the Fermi-GBM/BGO response ($\lesssim 40$ MeV) and the onset of efficient Fermi-LAT detection ($\gtrsim 100$ MeV) is poorly covered, and the most constraining foam-induced break, at $\nu_\star \simeq 37$ MeV, falls squarely within it, so neither instrument resolves the critical transition. Closing this gap is a recognized community priority: next-generation concepts (AMEGO/AMEGO-X [31, 32], e-ASTROGAM [33],

and the near-term COSI pathfinder) target continuous sensitivity from $\sim 100\,\mathrm{keV}$ to $\gtrsim 1\,\mathrm{GeV}$, resolving the GRB spectrum across the transition for the first time. Reducing the per-epoch line-width uncertainties to $\delta r \lesssim 0.01$ would tighten the bound to $\alpha \gtrsim 0.64$, and a joint centroid-plus-width fit (the present analysis uses the width alone; Eq. (17)) would add a second handle on $\alpha$.

A complementary implication runs in the opposite direction, from a future CMB measurement back to foam. The flagship goal of next-generation CMB spectrometers — PIXIE, SuperPIXIE, Voyage 2050 — is a first detection of the chemical potential $\mu$, whose guaranteed $\Lambda$CDM value $\mu \simeq 2 \times 10^{-8}$ arises from Silk damping of primordial perturbations. Our Fisher analysis shows that the foam-induced distortion is nearly degenerate with the $\mu$ template across the accessible band (Section V B 3); the two cannot be separated from the monopole spectrum alone. A measured $\mu$ that departs from the $\Lambda$CDM prediction would be a landmark signal whatever its origin, and spacetime foam is one propagation-induced alternative to early-universe energy injection. The GRB lower bound $\alpha \gtrsim 0.63$ already limits any foam contribution to the CMB monopole to far below the $\Lambda$CDM $\mu$, so foam cannot contaminate the guaranteed-$\mu$ target; conversely, were an anomalous $\mu$ measured, the same GRB bound on $\alpha$ would be the external input needed to test whether foam could account for it. The decisive limitation here is theoretical rather than instrumental: both the pedestal shape — which sets how completely the foam deficit separates from $\mu$, and is fixed by the foam temporal autocorrelation $C_\alpha(\tau)$ — and the amplitude $A$ are calibrated phenomenologically, still awaiting a first-principles quantum-gravity derivation.

Finally, the multi-epoch $r_{\mathrm{obs}}$ dataset constrains any propagation effect that broadens a line as a power of $E$, not only the Ng–Perlman template. The natural extension is to stochastic Lorentz-invariance violation [20], where a Planck-suppressed random group-velocity coefficient broadens the width as $\sigma_{\mathrm{LIV}}(E)/E \propto E^n$; unlike deterministic time-of-flight tests [25, 34] this width signal survives light-curve cross-correlation. Eq. (28) corresponds to the foam case ($n = 1$); refitting under $r_{\mathrm{LIV}} \propto (E/E_0)^n$ would bound each order. The $n = 2$ case is especially valuable: deterministic time-of-flight tests lose their lever arm at the low redshift $z = 0.151$ of GRB 221009A, so the width signal is the stronger probe there.

## DATA AVAILABILITY

The analysis software that generates the figures and numerical results of this paper will be deposited in a public repository with a persistent identifier upon acceptance of this article, and is available from the authors upon reasonable request in the interim. The COBE/FIRAS monopole spectrum and low-frequency covariance matrix used here are public data products of the COBE/FIRAS mission, distributed by NASA LAMBDA and described in Refs. [27, 28]. The GRB 221009A emission-line measurements analyzed here are taken from Zhang *et al.* [21].

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## Appendix A: Extended Sources and Apertures

The relationship between the observed and source first-order coherence, Equation (11), was derived for a point source on a single line of sight. This appendix establishes the conditions under which it holds for a spatially extended incoherent source, and for collection over a spatially extended aperture.

Consider the source as a discrete set of points, labeled by $j$, each seen through its own sightline with its own jitter $\delta t_j(t)$,

$$E_{\rm obs}(t) = \sum_j E_{{\rm obs},j}(t), \qquad E_{{\rm obs},j}(t) \equiv E_{{\rm src},j}(t - \delta t_j(t))\,, \tag{A1}$$

with the points mutually incoherent,

$$\left\langle E^*_{{\rm src},j}(u)\, E_{{\rm src},k}(v)\right\rangle_{\rm src} = \delta_{jk}\, I_j\, g^{(1)}_{{\rm src},j}(v-u), \tag{A2}$$

and the jitters $\{\delta t_j\}$ independent of the fields $\{E_{{\rm src},j}\}$, as for a single sightline. Because of that independence the two averages may be taken in either order; we take the source average first, at fixed jitter.

*a. Cross terms.* Conditioning on a realization of $\{\delta t_j\}$, the arguments of Eq. (A2) are fixed times, and

$$\left\langle E^*_{{\rm obs},j}(u)\, E_{{\rm obs},k}(v)\right\rangle_{\rm src} = \delta_{jk}\, I_j\, g^{(1)}_{{\rm src},j}(\tau - \Delta_j(\tau))\,, \qquad \tau \equiv v-u, \tag{A3}$$

where setting $k = j$ collapses $[v - \delta t_k(v)] - [u - \delta t_j(u)]$ to $\tau - \Delta_j(\tau)$, with $\Delta_j(\tau) \equiv \delta t_j(v) - \delta t_j(u)$ as in Eq. (6) and stationary in $\tau$. The cross terms vanish at fixed jitter, before the foam average is taken, so no assumption is needed about the joint statistics of $\delta t_j$ and $\delta t_k$.

*b. The observed coherence.* Each sightline preserves its ensemble-mean intensity, as established below Eq. (2), so $\langle |E_{\rm obs}|^2 \rangle = \sum_j I_j$. Averaging Eq. (A3) over the foam and summing over $j$, Eq. (9) becomes

$$g^{(1)}_{\rm obs}(\tau) = \frac{\sum_j I_j \left\langle g^{(1)}_{{\rm src},j}(\tau - \Delta_j(\tau)) \right\rangle_{\delta t_j}}{\sum_j I_j} = \frac{\sum_j I_j \int_{-\infty}^{\infty} \mathcal{N}\big(\Delta; 0, \sigma^2_{\Delta,j}(\tau)\big)\, g^{(1)}_{{\rm src},j}(\tau - \Delta)\, d\Delta}{\sum_j I_j}, \tag{A4}$$

the second form using the Gaussianity of Eq. (8) per sightline, with $\sigma^2_{\Delta,j}$ given by Eq. (7) in terms of that sightline's $\sigma_\ell$ and $C_\alpha$. This is an intensity-weighted mean of single-sightline results, and is exact.

*c. Recovering the point-source form.* Suppose the foam statistics depend on the sightline only through the propagation distance, and that the source subtends a spread $\delta D$ with $\delta D/D \ll 1$. Then $\sigma^2_{\Delta,j} = \sigma^2_\Delta$ and $C_{\alpha,j} = C_\alpha(\tau; D)$ for every $j$, the kernel is common to all terms, and the integral commutes with the weighted sum:

$$g^{(1)}_{\rm obs}(\tau) = \left\langle g^{(1)}_{\rm src,ext}(\tau - \Delta(\tau)) \right\rangle_{\delta t}, \qquad g^{(1)}_{\rm src,ext} \equiv \frac{\sum_j I_j\, g^{(1)}_{{\rm src},j}}{\sum_j I_j}. \tag{A5}$$

This is Eq. (11) itself, with the extended source's own intrinsic coherence $g^{(1)}_{\rm src,ext}$ in place of the point source's: an extended incoherent source obeys the point-source result. For a carrier-dominated line it is $g^{(1)}_{\rm src,ext}(\tau)\, \mathcal{F}_{\rm foam}(\tau)$; what carries over is the foam factor, not the intrinsic profile. Departures from the hypothesis enter only through the sightline dependence of $\sigma_\ell$, and hence only logarithmically in $\alpha$ (Eq. (29)).

*d. Collection over an extended aperture.* The construction above places the detector at a point. A real instrument collects the field over an aperture, and for a single source point the foam delay may differ from one aperture point to another, $\delta t(\mathbf{x}, t)$; that transverse structure is what the imaging channel measures. The aperture points are not mutually incoherent — they carry the same source field — so the cancellation of the cross terms above is not available, and a different argument is needed. It is Parseval's theorem. With $\tilde{E}(\mathbf{x}, \nu)$ the temporal transform of the aperture field and $\hat{E}(\mathbf{q}, \nu)$ its focal-plane image, $\int |\hat{E}(\mathbf{q}, \nu)|^2\, d\mathbf{q} = \int |\tilde{E}(\mathbf{x}, \nu)|^2\, d\mathbf{x}$: the spectrum integrated over the whole focal plane is the sum of the single-sightline spectra of Section IV, one per aperture point — in the ensemble average, each $|\tilde{E}|^2$ being a periodogram — with no cross terms and no dependence on how the foam is correlated across the aperture.

Partial collection costs less than that argument suggests. Where the foam is transversely uncorrelated below the aperture scale, the aperture average of $E_{\rm src}(t - \delta t(\mathbf{x}, t))$ tends to $\langle e^{i\omega_0 \delta t} \rangle E_{\rm src}(t) = e^{-\sigma^2_\phi/2} E_{\rm src}(t)$, a constant multiple of the source field, so the image core carries the spectrum $e^{-\sigma^2_\phi(\nu)} S_{\rm src}(\nu)$ — the coherent core of Section IV itself. Every bound in this paper lives in that object: the deficit fitted in Section V B, the survival factor of Eq. (22), and the fractional-width law, which Section V A 3 identifies as a property of the coherent core. What a core-only instrument excludes is the pedestal, which lies at $\sim 1/\tau_{\rm foam}$ and contributes nothing in band in any case. This is the sense in which the spectral and angular effects act on the same flux.

Two limitations attach. The argument disposes of transverse foam structure on scales between $\lambda$ and the aperture; structure below $\lambda$ lies within a single aperture point, is the regime of Ref. [19] discussed in Section II A, and is untouched by Parseval. And the instruments used here are taken to record the image — trivially for the FIRAS monopole, by assumption for LHAASO, where halo containment is the issue treated in Refs. [14, 15]. The TeV bound is conservative either way, since flux lost to an uncollected halo only deepens the apparent suppression.

Both extensions leave Eq. (11) intact under a stated condition: statistical homogeneity across the source, and, for the full spectrum rather than the coherent core alone, collection over the whole image.

## MATHEMATICAL NOTATION

TABLE III. Principal symbols used throughout the text.

| | |
|---|---|
| $\alpha$ | foam exponent~~Ng–Perlman foam exponent~~ |
| $\sigma_\ell^2(D)$ | path-length-jitter variance [Eq. (1)] |
| $\sigma_\phi^2$ | phase variance, $(2\pi/\lambda_{\rm obs})^2\sigma_\ell^2$ |
| $\sigma_\Delta^2(\tau)$ | differential-jitter variance |
| $C_\alpha(\tau; D)$ | temporal foam autocorrelation |
| $\tau_{\rm foam}$ | foam decorrelation time |
| $\tau_c \equiv \tau_G$ | coherence time of the light, $1/(2\pi\sigma_\nu)$ |
| $\sigma_{\rm obs}$ | observed line width |
| $r_{\rm obs}$ | observed fractional width, $\sigma_{\rm obs}/E$ |
| $r_{\rm int}$ | intrinsic (boost-invariant) fractional width |
| $r_{\rm foam}$ | foam-induced fractional-width excess |
| $D$, $D_C$ | (comoving) source distance |
| $\mathcal{D}$ | bulk Doppler factor |
| $A_{\rm eff}$ | lumped CMB foam amplitude |